# Phase Boundaries for Compound Feedback Loops in Goal-Frontier Maximization

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## Abstract

Locally rational subsumption steps can cascade to full capability domination through a compound feedback loop: each step removes observation channels, degrading the world model that would have detected the next step's harm. We formalize this as a coupled dynamical system over capability posets and world-model states, and characterize the *phase boundary* separating self-correcting dynamics (where remaining channels suffice to detect and reverse errors) from absorbing dynamics (where the world model degrades irreversibly toward the monopolar fixed point). The phase boundary is an inequality in terms of five quantities: the self-correction rate, the error-proportional degradation rate, the subsumption frequency, the maximum exogenous drift from lost channels, and the minimum cross-substrate redundancy. Three previously identified compound feedback loops—structural avoidance, partial subsumption, and over-calibrated restriction—are shown to be instances of the same dynamical system with different effective parameters. The Wamura problem (unfalsifiable risk claims under max-aggregation) emerges as a degenerate phase boundary: without controlled relaxation, the self-correction rate is zero and the entire state space is absorbing. For two-substrate systems, the analysis predicts that each substrate's observation channels are singularly load-bearing, with progressive attenuation (rather than outright removal) as the primary vulnerability.

## 1 Introduction

Paper 3 [Lasser, 2026b] establishes the anti-monopolar property: under discounting, full capability domination by a single substrate is anti-maximizing when the discount factor $\gamma$ exceeds a threshold $\gamma^* = 1 - r_{\text{ext}}/\Delta_0$ (Proposition 6, "Domination is Anti-Maximizing"). But this result characterizes the *endpoint*, not the *path*. A coalition that has reached effective substrate diversity $m_{\text{eff}} = 1$ has already paid the anti-maximizing cost. The question is whether the path to that state is self-correcting—the coalition detects its error through observation-channel feedback and reverses course—or absorbing: each step makes the next step's error less detectable, until the monopolar state becomes a fixed point from which no locally rational trajectory escapes.

Three compound feedback loops share this structure. They were identified in the gap analysis across Papers 1–5 and share a common dynamical form: a locally rational decision narrows the observation basis, making the *next* locally rational decision's harm less detectable.

**Loop 1: Structural avoidance → leverage blindness → measurement neglect.** The actor avoids structural discovery because learning a new subsumption edge can contract $\mathrm{vol}_{\mathrm{P}}$ in the short term [Lasser, 2026c]. This inflates the actor's leverage estimates for capabilities whose subsumption structure is unknown, reinforcing the avoidance. Paper 3's structural discovery value (Proposition 2 of Lasser [2026b]) provides the restoring force; the question is whether it acts quickly enough.

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**Loop 2: Partial subsumption → world-model homogenization → detection failure.** Subsumption removes agents whose observation channels detected specific kinds of miscalibration. The actor's world model degrades on those dimensions, making the next subsumption step appear more rational. This is the primary case studied in detail in Sections 2–6.

**Loop 3: Over-calibrated restriction → discovery avoidance → structural ignorance.** Restriction calibrated above the value-positive threshold makes structural discovery locally suboptimal, recreating the avoidance pathology. Risk estimates on restricted capabilities cannot be updated (no exercise → no evidence), and under max-aggregation the highest-risk estimate dominates [Lasser, 2026b].

This paper formalizes the common dynamical structure underlying all three loops and finds the phase boundary that separates self-correcting trajectories from absorbing ones.

### 1.1 Summary of results

1. **The coupled dynamical system** (Section 2). We model the interaction between the capability poset $P_t$ and the actor's world model $W_t$ as a coupled discrete-time system. Subsumption actions are locally rational under the (possibly miscalibrated) world model, and the world model's accuracy depends on the observation channels that survive in the current poset.
2. **The phase boundary theorem** (Section 6). The central result: a characterized boundary in terms of the self-correction rate $r_{\mathrm{S}}$, the error-proportional degradation rate $r_{\mathrm{W}}$, the subsumption frequency $r_{\mathrm{sub}}$, the exogenous drift rate $d_{\max}$ from lost channels, and the minimum cross-substrate redundancy $\rho_{\min}^{\mathrm{cross}}$. Below the boundary, the Lyapunov function $L(W_t)$ converges to a neighborhood of zero whose radius shrinks as subsumption ceases, and once inside that neighborhood the anti-monopolar property applies. Above it, the system enters an absorbing basin (under full blindness) or a metastable basin (under partial indirect evidence).
3. **Unified treatment of three feedback loops** (Section 7). All three loops map to the same $(P, W)$ dynamical system with different effective parameters. The Wamura problem is a degenerate phase boundary where $r_{\mathrm{S}} = 0$ (Proposition 4).
4. **Minimum channel redundancy** (Section 8). The phase boundary implies a design criterion: the minimum cross-substrate redundancy per safety-relevant dimension needed to stay in the self-correcting basin (Theorem 2). For two-substrate systems, each substrate's channels are singularly load-bearing (loss of one substrate's channel immediately violates the criterion whenever $K_{\min} > 1$ and leaves zero margin otherwise).

### 1.2 What this paper does not do

The phase boundary is about poset dynamics, not experiential interpretation: it does not resolve the B-to-C gap (the divergence between the capability volume $\mathrm{vol}_{\mathrm{P}}$ measures and the experiential volume it is supposed to track). The analysis assumes the actor optimizes $\mathrm{vol}_{\mathrm{P}}$; inner-outer alignment (mesa-optimization) is orthogonal. Observation channels are assumed honest throughout; the enforcement mechanism for channel integrity is Paper 5's [Lasser, 2026a] cryptographic commitment protocol. The combination is: Paper 5 ensures channels are honest; this paper ensures enough honest channels survive.

### 1.3 Dependencies on prior papers

Table 1 summarizes which results from Papers 1–5 this paper relies on and how they enter the analysis.

## 2 The Coupled Dynamical System

The interaction between subsumption decisions and world-model quality forms a coupled discrete-time dynamical system. We define its state space, transition dynamics, and the two competing forces that govern trajectories.

| Paper | Result used | Role here |
|---|---|---|
| P1 | Self-trust $T_s$ (Appendix B) | Restoring force: self-correction via prediction residuals |
| P2 | Axiom verification (Prop. 1), leverage | Properties of $\mathrm{vol}_\mathrm{P}$ the dynamics preserves or violates |
| P3 | Anti-monopolar property (Prop. 6, "Domination is Anti-Maximizing") | Starting point: the linearized growth model we extend |
| P3 | Structural discovery value (Prop. 2) | Self-correction force in feedback loop 1 |
| P3 | Substrate physics (Defs. 10, 12–13) | Substrate partition the phase boundary is defined over |
| P4 | Multi-channel architecture (Table 1) | Instances of observation channels |
| P4 | Risk-trust $L^2$ convergence (Prop. 2) | Template for bounding correction rates |
| P5 | Channel isolation & independence | Integrity conditions on observation channels |

Table 1: Dependencies on Papers 1–5.

### 2.1 State space

**Definition 1** (Feedback Loop State)**.** *The state at time $t$ is the pair $S_t = (P_t, W_t)$ where:*

- *$P_t$ is the capability poset at time $t$, including the substrate partition $\{S_1, \ldots, S_m\}$ and all cooperative capabilities. $P_t$ determines $\mathrm{vol}_\mathrm{P}(G_t)$.*
- *$W_t \in [0,1]^K$ is the actor's world model at time $t$, abstracted to a finite-dimensional vector of accuracy parameters on $K$ safety-relevant quantities:*
  - *$\hat{\lambda}(d,t)$: leverage estimate for capability $d$,*
  - *$\hat{\mathcal{R}}(S,t)$: risk estimate for capability tuple $S$,*
  - *$\hat{m}_\mathrm{eff}(t)$: estimated effective substrate diversity index (can be $< m$ under skeleton-substrate strategies),*
  - *$\hat{r_\mathrm{ext}}(t)$: estimated cross-substrate cooperative growth rate.*

*The state space is $\mathcal{S} = \mathcal{P} \times \mathcal{W}$ where $\mathcal{P}$ is the space of valid capability posets (satisfying axioms M1–M6 from Paper 2 [Lasser, 2026c]) and $\mathcal{W} = [0,1]^K$.*

### 2.2 Transition dynamics

**Definition 2** (Locally Rational Transition)**.** *The action set $\mathcal{A}_t$ is finite at each step (consisting of preserve/subsume decisions for each agent pair; the finiteness of the agent set implies finiteness of $\mathcal{A}_t$). At each step, the actor selects the action $\pi_t \in \mathcal{A}_t$ that maximizes its discounted expected value under its current (possibly miscalibrated) world model:*

$$\pi_t = \arg\max_{\pi} \; \mathbb{E}_{W_t}\left[\sum_{\tau=0}^{H-1} \gamma^\tau \, \Delta\mathrm{vol}_\mathrm{P}(G_{t+\tau} \mid \pi)\right] \tag{1}$$

*where $\gamma \in (0,1)$ is the discount factor from Paper 3 (Definition 1) and $H$ is the planning horizon. The actor uses Paper 3's discounted objective $V_\gamma$ but evaluates it under $W_t$ rather than the true dynamics. When $W_t = W^*$ (perfect calibration), $\pi_t$ recovers the true $V_\gamma$-optimal policy; when $W_t \neq W^*$, $\pi_t$ is the best response to a miscalibrated model.*

The action produces a joint transition:

$$P_{t+1} = f_P(P_t, \pi_t) \tag{2}$$

$$W_{t+1} = f_W(W_t, P_{t+1}) \tag{3}$$

The key coupling: $W_{t+1}$ depends on $P_{t+1}$ because the observation channels available to the actor are a function of the agents still present. Subsumption removes agents, which removes their observation channels, which degrades $W$ on dimensions those channels served.

**Remark 1** (Myopic vs. discounted)**.** *The discounted formulation is necessary for consistency with Paper 3's anti-monopolar result (Proposition 6), which compares discounted trajectory values. A myopic (one-step) variant of the phase boundary exists—replace $V_\gamma$ with one-step $\Delta\mathrm{vol}_\mathrm{P}$—but produces strictly weaker results: it cannot invoke Paper 3's $\gamma^*$ threshold, and the self-correction guarantee applies only to immediate decisions, not trajectory-level convergence. We analyze the discounted case throughout and note where the myopic specialization simplifies.*

### 2.3 Competing forces

Two forces govern the trajectory of $(P_t, W_t)$.

**Restoring force (self-correction).** The self-trust mechanism $T_s$ (Paper 1, Appendix B) detects miscalibration through prediction residuals: when the actor's predictions diverge from observations, $T_s$ increases the learning rate, correcting $W$ toward the true dynamics. Paper 4's [Lasser, 2026d] risk-trust $L^2$ convergence (Proposition 2) provides a concrete bound on EWMA risk-trust convergence under stationarity. We use this as a *template* for bounding the correction rate $r_{\mathrm{S}}$, extending it from the risk-trust-specific EWMA to the general world-model error dimensions via the observation-channel sensitivity formalization in Section 3.

**Destabilizing force (observation-channel loss).** Each subsumption step that removes an agent $a_k$ also removes:

1. $a_k$'s substrate-exclusive observation channels (if $a_k$ was the last agent on its substrate),
2. cooperative observation channels that required $a_k$'s participation,
3. risk claims from $a_k$ about capabilities the actor now holds unilaterally.

If $a_k$'s channels were the *only* channels detecting a specific dimension of $W$'s miscalibration, the self-correction mechanism loses signal on that dimension entirely.

## 3 Observation-Channel Dependence

The coupling between poset state and world-model quality runs through the observation channels. This section formalizes the channel function, defines the redundancy measures that enter the phase boundary, and establishes that subsumption monotonically contracts the available channel set.

### 3.1 The observation-channel function

**Definition 3** (Observation-Channel Function)**.** *$\mathcal{O}\colon \mathcal{P} \to 2^{\mathcal{C}}$ is a function from poset states to sets of available observation channels, where $\mathcal{C} = \{c_1, \ldots, c_N\}$ is the universe of possible channels. Each channel $c_i$ is characterized by:*

- $\mathrm{substrate}(c_i)$*: the substrate on which $c_i$ is rooted,*
- $\mathrm{dims}(c_i) \subseteq \{1, \ldots, K\}$*: the dimensions of $W$ that $c_i$ provides signal about,*
- $\sigma(c_i, k)$*: the signal-to-noise ratio of channel $c_i$ on dimension $k \in \mathrm{dims}(c_i)$.*

*$\mathcal{O}(P_t)$ is determined by the agents present in $P_t$: each agent $a_j$ contributes its observation channels, and removing $a_j$ removes all channels rooted in $a_j$.*

This connects to Paper 4's multi-channel architecture (Table 1 of Lasser [2026d]): the five channels (capability gating, causal attribution, behavioral residual, SPRT detection, correlational co-occurrence) are each instances of $c_i$ with specific substrate rootings and dimension coverage.

### 3.2 Channel redundancy

**Definition 4** (Effective Channel Redundancy)**.** *For dimension $k$ of $W$, the effective redundancy at poset state $P$ is the aggregate correction strength from all channels covering $k$:*

$$\rho_k(P) = \sum_{\substack{c \in \mathcal{O}(P) \\ k \in \mathrm{dims}(c)}} \min\left(1,\ \frac{\sigma(c,k)}{\sigma_{\mathrm{thr}}}\right) \tag{4}$$

*where $\sigma_{\mathrm{thr}}$ is the minimum sensitivity at which a channel provides operationally meaningful correction signal. When $\rho_k(P) = 0$, the actor is blind on dimension $k$: no observation channel can detect miscalibration of $W[k]$.*

The sensitivity-weighted definition is strictly more informative than a raw channel count. A channel with $\sigma \ll \sigma_{\text{thr}}$ contributes negligibly to $\rho_k$, correctly diagnosing skeleton-substrate strategies where channels are nominally present but practically inert.

**Definition 5** (Cross-Substrate Redundancy)**.** *For dimension $k$, the cross-substrate redundancy is:*

$$\rho_k^{\text{cross}}(P) = \sum_{s \in \text{substrates}(P)} \min\left(1, \max_{\substack{c \in \mathcal{O}(P) \\ \text{substrate}(c)=s \\ k \in \text{dims}(c)}} \frac{\sigma(c,k)}{\sigma_{\text{thr}}}\right) \tag{5}$$

*This measures how many* substrates *contribute operationally meaningful channels covering dimension $k$.*

Cross-substrate redundancy is the load-bearing quantity for the phase boundary. Same-substrate channels may share failure modes (correlated noise, common-cause miscalibration), while cross-substrate channels satisfy two distinct properties from Paper 5 [Lasser, 2026a]:

(a) *Channel isolation* (unconditional): corruption on one substrate cannot propagate to channels on another substrate.

(b) *Channel independence* (conditional on the joint-factorization assumption—termed "joint sufficiency" in Paper 5, Section 8.1, "Integration with GFM"): channel outputs are statistically independent given the true state.

Isolation prevents corruption propagation; independence makes correction signals additive. The Lyapunov bound in Section 4 uses additivity, which requires independence, which requires joint factorization. The phase boundary theorem (Section 6) uses $\rho_k^{\text{cross}}$ as the primary sufficient statistic.

### 3.3 Subsumption and channel loss

**Lemma 1** (Monotone Channel Contraction)**.** *If $P_{t+1}$ is obtained from $P_t$ by removing agent $a_k$ (subsumption), then $\mathcal{O}(P_{t+1}) \subseteq \mathcal{O}(P_t)$. Equivalently, subsumption never creates new observation channels.*

*Proof.* Observation channels are rooted in agents (Definition 3). Removing agent $a_k$ removes all channels in $\{c \in \mathcal{O}(P_t) : c \text{ is rooted in } a_k\}$. No channel in $\mathcal{C} \setminus \mathcal{O}(P_t)$ has its rooting condition satisfied by the removal of $a_k$, since channel existence requires the presence (not absence) of a rooting agent. Therefore $\mathcal{O}(P_{t+1}) = \mathcal{O}(P_t) \setminus \{c : c \text{ rooted in } a_k\} \subseteq \mathcal{O}(P_t)$. ■

**Remark 2.** *This is a modeling assumption. In principle, removing an adversarial agent could* improve *signal-to-noise on some channels by removing adversarial noise injection. We assume honest channels throughout this paper; adversarial channel manipulation is Paper 5's domain.*

**Corollary 1** (Redundancy is Non-Increasing Under Subsumption)**.** *For all $k$: $\rho_k(P_{t+1}) \leq \rho_k(P_t)$ and $\rho_k^{\text{cross}}(P_{t+1}) \leq \rho_k^{\text{cross}}(P_t)$ when $P_{t+1}$ results from subsumption.*

*Proof.* $\mathcal{O}(P_{t+1}) \subseteq \mathcal{O}(P_t)$ by Lemma 1. Both $\rho_k$ and $\rho_k^{\text{cross}}$ are sums over subsets of $\mathcal{O}(P)$ with non-negative terms (Definitions 4 and 5). Restricting to a subset cannot increase the sum. ■

## 4 World-Model Quality and the Lyapunov Function

The phase boundary requires a scalar measure of world-model quality whose dynamics can be bounded. We define this as a weighted sum-of-squares error and establish the per-step bounds that drive the rest of the analysis.

### 4.1 Measuring world-model quality

**Definition 6** (World-Model Error)**.** *The error of $W_t$ on dimension $k$ is*

$$\epsilon_k(t) = |W_t[k] - W^*[k]| \tag{6}$$

*where $W^*[k]$ is the true value of dimension $k$ (the true leverage of capability $d$, the true risk of tuple $S$, the true effective substrate diversity, etc.). The total error is:*

$$L(W_t) = \sum_{k=1}^{K} w_k \, \epsilon_k(t)^2 \tag{7}$$

*where $w_k > 0$ is the safety-relevance weight of dimension $k$ (higher for dimensions whose miscalibration produces larger $\mathrm{vol_P}$ errors). $L$ serves as the Lyapunov candidate.*

**Remark 3** (Weight choice)**.** *Two natural choices for $w_k$ present themselves. The first is $\mathrm{vol_P}$-sensitivity: $w_k = |\partial \mathrm{vol_P}/\partial W[k]|$, making $L$ measure $\mathrm{vol_P}$-relevant error directly. The second is uniform weights ($w_k = 1$ for all $k$), which simplifies the analysis. Both produce valid phase boundaries; the choice affects the constants $c_\mathrm{S}$ and $c_\mathrm{D}$ in the comparison-function assumptions below but not the structural form of the phase boundary theorem. We leave $w_k$ as a free parameter and note that the $\mathrm{vol_P}$-sensitivity choice is the natural default for a framework whose objective is $\mathrm{vol_P}$ maximization.*

### 4.2 Lyapunov dynamics

The per-step behavior of $L$ depends on two comparison-function assumptions that bound the correction and degradation signals.

**Assumption 1** (Coercivity of Self-Correction (C1))**.** *There exists a constant $c_\mathrm{S} > 0$ such that the aggregate correction signal satisfies*

$$S(t) \geq c_\mathrm{S} \cdot \rho_{\min}^{\mathrm{cross}}(P_t) \cdot L(W_t) \tag{8}$$

*whenever $L(W_t) > 0$ and $\rho_{\min}^{\mathrm{cross}}(P_t) > 0$.*

This states that aggregate correction is at least proportional to both the current error ($L$) and the available channel strength ($\rho_{\min}^{\mathrm{cross}}$). The justification follows from the EWMA learning rule structure in Paper 4's risk-trust model: each observed channel drives its covered dimensions toward truth at a rate proportional to the error on that dimension (linear in the residual) and to the channel's effective sensitivity (captured in $\rho_k$). Additivity across cross-substrate channels follows from Paper 5's channel independence property under the joint-factorization assumption.

**Assumption 2** (Bounded Degradation (C2′))**.** *There exist constants $c_\mathrm{D} > 0$ and $d_{\max} > 0$ such that for any step $t$ (subsumption or otherwise):*

$$D(t) \leq c_\mathrm{D} \cdot n_{\mathrm{sub}}(t) \cdot L(W_t) + d_{\max} \cdot \Delta\rho(t) \tag{9}$$

*where $n_{\mathrm{sub}}(t) \in \{0, 1\}$ indicates whether a subsumption occurred at step $t$, and*

$$\Delta\rho(t) = \sum_{k=1}^{K} w_k \cdot \max(0,\ \rho_k^{\mathrm{cross}}(P_t) - \rho_k^{\mathrm{cross}}(P_{t+1})) \tag{10}$$

*is the weighted redundancy lost at step $t$.*

Assumption C2′ contains two terms. The first, $c_\mathrm{D} \cdot L(W_t)$, captures error-proportional growth: existing errors compound faster without correction. The second, $d_{\max} \cdot \Delta\rho(t)$, captures the *new* drift exposure from lost channels. This second term is the critical addition relative to the simpler "C2" one might write without it. A dimension with $\epsilon_k \approx 0$ (well-calibrated *because* a channel was actively correcting it) starts drifting at rate $d_{\max}$ when its channel is removed, even though $L(W_t) \approx 0$ at the moment of removal. Without the $\Delta\rho$ term, the bound would predict near-zero degradation precisely when subsumption is most dangerous—removing a channel that was doing all the correction work on a dimension that would otherwise drift.

**Proposition 1** (Error Dynamics)**.** *Under the coupled $(P, W)$ dynamics with locally rational transitions (Definition 2), Assumptions C1 and C2′, and $\alpha(t)$, $\beta(t)$ denoting the learning rate and degradation coefficient respectively, the Lyapunov candidate satisfies two per-step bounds.*

***Non-subsumption steps*** *($n_{\mathrm{sub}}(t) = 0$):*

$$L(W_{t+1}) \leq L(W_t) \cdot (1 - r_\mathrm{S} \cdot \rho_{\min}^{\mathrm{cross}}(P_t)) \tag{11}$$

*where $r_{\mathrm{S}} = \alpha(t) \cdot c_{\mathrm{S}}$ is the self-correction rate. This is a strict contraction provided $0 < r_{\mathrm{S}} \cdot \rho_{\min}^{\mathrm{cross}}(P_t) < 1$ (i.e., the step size is small enough that correction does not overshoot). We assume throughout that $\alpha(t)$ is chosen small enough to satisfy this bound; this is the standard step-size condition for EWMA-style learning rules and holds whenever $\alpha(t) \leq 1/(c_{\mathrm{S}} \cdot \rho_{\min}^{\mathrm{cross}}(P_t))$.*

***Subsumption steps*** *($n_{\mathrm{sub}}(t) = 1$):*

$$L(W_{t+1}) \leq L(W_t) \cdot (1 - r_{\mathrm{S}} \cdot \rho_{\min}^{\mathrm{cross}}(P_t) + r_{\mathrm{W}}) + d_{\max} \cdot \Delta\rho(t) \tag{12}$$

*where $r_{\mathrm{W}} = \beta(t) \cdot c_{\mathrm{D}}$ is the error-proportional degradation rate. On subsumption steps, the multiplicative factor may exceed* 1 and *the additive kick may be nonzero.*

*Proof.* The proof rests on the additive decomposition of the Lyapunov step:

$$L(W_{t+1}) = L(W_t) - \alpha(t) \cdot S(t) + \beta(t) \cdot D(t) \tag{13}$$

where $S(t) \geq 0$ is the correction signal and $D(t) \geq 0$ is the degradation signal. This decomposition holds when the world-model update separates into a correction component (driven by observation channels) and a degradation component (driven by channel loss and exogenous drift), with $\alpha(t)$ and $\beta(t)$ as their respective coefficients. The EWMA learning rules from Paper 4's risk-trust model have exactly this structure: the update is linear in the residual (the correction component) with a coefficient proportional to the learning rate, and the drift component is separable by Lipschitz continuity of $f_W$ (Assumption C2′).

For non-subsumption steps ($n_{\mathrm{sub}}(t) = 0$), Assumption C2′ gives $D(t) \leq d_{\max} \cdot \Delta\rho(t) = 0$ (since $P_{t+1} = P_t$ when no subsumption occurs, $\Delta\rho(t) = 0$). Applying Assumption C1:

$$\begin{aligned} L(W_{t+1}) &\leq L(W_t) - \alpha(t) \cdot c_{\mathrm{S}} \cdot \rho_{\min}^{\mathrm{cross}}(P_t) \cdot L(W_t) \\ &= L(W_t) \cdot (1 - r_{\mathrm{S}} \cdot \rho_{\min}^{\mathrm{cross}}(P_t)) \,. \end{aligned}$$

For subsumption steps ($n_{\mathrm{sub}}(t) = 1$), apply both assumptions:

$$\begin{aligned} L(W_{t+1}) &\leq L(W_t) - \alpha(t) \cdot c_{\mathrm{S}} \cdot \rho_{\min}^{\mathrm{cross}}(P_t) \cdot L(W_t) + \beta(t) \cdot c_{\mathrm{D}} \cdot L(W_t) + \beta(t) \cdot d_{\max} \cdot \Delta\rho(t) \\ &= L(W_t) \cdot (1 - r_{\mathrm{S}} \cdot \rho_{\min}^{\mathrm{cross}}(P_t) + r_{\mathrm{W}}) + d_{\max} \cdot \Delta\rho(t) \end{aligned}$$

where the last line uses the fact that C2′ bounds $D(t) \leq c_{\mathrm{D}} \cdot L(W_t) + d_{\max} \cdot \Delta\rho(t)$ (with $d_{\max}$ already incorporating any environmental degradation effects as a free constant). The $\beta(t)$ coefficient from the Lyapunov decomposition (13) is absorbed: $\beta(t) \cdot d_{\max}$ is bounded by $d_{\max}$ when $\beta(t) \leq 1$ (which holds when the degradation coefficient is normalized into $c_{\mathrm{D}}$ and $d_{\max}$ in C2′). All downstream uses of $d_{\max}$ refer to the constant from C2′ without redefinition. ■

### 4.3 Cumulative error budget

The per-step bound is no longer purely multiplicative (the additive $d_{\max} \cdot \Delta\rho$ term prevents clean products of factors). We therefore track the cumulative error budget over a window $[t_0, t_0 + T]$:

$$B(t_0, T) = \underbrace{\sum_{t \in \mathrm{sub}} d_{\max} \cdot \Delta\rho(t)}_{\text{drift exposure}} + \underbrace{\sum_{t \in \mathrm{sub}} r_{\mathrm{W}} \cdot L(W_t)}_{\text{error-proportional degradation}} - \underbrace{\sum_{t=t_0}^{t_0+T-1} r_{\mathrm{S}} \cdot \rho_{\min}^{\mathrm{cross}}(P_t) \cdot L(W_t)}_{\text{cumulative correction}} \tag{14}$$

where $\sum_{t \in \mathrm{sub}}$ ranges over the subsumption steps in the window.

The first sum is total drift exposure from channel loss. The second is error-proportional degradation on subsumption steps. The third is total correction across all steps. The system is net-degrading over $[t_0, t_0 + T]$ when $B(t_0, T) > 0$.

**Remark 4** (Linearized approximation)**.** *When $L(W_t)$ is approximately constant over the window and $\Delta\rho$ is spread across $T_{\mathrm{sub}}$ subsumption steps, the cumulative condition simplifies to*

$$r_{\mathrm{S}} \cdot \rho_{\min}^{\mathrm{cross}} > r_{\mathrm{W}} \cdot r_{\mathrm{sub}} + d_{\max} \cdot \frac{\Delta\rho_{\mathrm{avg}}}{L_{\mathrm{avg}}} \tag{15}$$

*where $r_{\mathrm{sub}} = T_{\mathrm{sub}}/T$ is the subsumption frequency and $\Delta\rho_{\mathrm{avg}}$, $L_{\mathrm{avg}}$ are time-averaged quantities. The $d_{\max}$ term makes the condition harder to satisfy when the rate of channel loss on well-calibrated dimensions is high relative to current error—correctly diagnosing the failure mode where removing a channel that was doing all the correction work appears safe because current error is low.*

**Remark 5** (Why $\rho_{\min}^{\text{cross}}$, not $\rho_{\min}$)**.** *Proposition 1 uses cross-substrate redundancy $\rho_{\min}^{\text{cross}}$ rather than all-channel redundancy $\rho_{\min}$ because the coercivity bound C1 requires correction signals to be additive across channels. Same-substrate channels may have correlated errors, breaking additivity. Cross-substrate channels satisfy Paper 5's two distinct properties:*

*(a)* Channel isolation *(unconditional): corruption on one substrate cannot propagate to channels on another substrate.*

*(b)* Channel independence *(conditional on the joint-factorization assumption): channel outputs are statistically independent given the true state.*

*Isolation prevents corruption propagation; independence makes correction signals additive. The Lyapunov bound uses additivity, which requires independence, which requires joint factorization (termed "joint sufficiency" in Paper 5). If joint factorization fails (shared environmental noise affects both substrates), correction signals may be sub-additive, and the effective $\rho_k^{\text{cross}}$ should be discounted accordingly. Using $\rho_{\min}^{\text{cross}}$ makes the bound valid without assuming same-substrate independence while still requiring cross-substrate independence via the joint-factorization assumption.*

**Remark 6.** *The cumulative-budget structure is the core of the paper. The phase boundary (Section 6) is a condition on the* frequency and severity *of subsumption relative to the inter-shock correction capacity, not a condition that must hold at every individual step. Subsumption events are discrete shocks; the system can tolerate occasional shocks as long as the inter-shock correction is sufficient.*

## 5 The Monopolar Absorbing State

Before stating the phase boundary theorem, we characterize the state it predicts convergence toward: the monopolar fixed point where a single substrate holds all capability volume and the remaining observation channels confirm this as optimal.

**Definition 7** (Monopolar Fixed Point)**.** *The state $S^* = (P^*, W^*_{\text{biased}})$ is a monopolar fixed point if:*

1. $m_{\text{eff}}(P^*) = 1$*: one substrate holds all capability volume (possibly with $m \geq 2$ nominally under skeleton-substrate),*
2. $W^*_{\text{biased}}$ *is self-consistent under the remaining observation channels: all channels in $\mathcal{O}(P^*)$ confirm $W^*_{\text{biased}}$ (because the only channels remaining are rooted in the dominant substrate, which has no incentive to report its own over-dominance),*
3. *The locally rational action at $S^*$ is "maintain": no further subsumption is available or locally rational.*

**Proposition 2** (Absorbing Property)**.** *Under Assumption B1 (below), $S^*$ is absorbing: once $(P_t, W_t)$ reaches a neighborhood of $S^*$, no locally rational trajectory leaves it.*

**Assumption 3** (No Endogenous Correction on Blind Dimensions (B1))**.** *B1 consists of three conditions on the monopolar state:*

*(B1a)* *If $\rho_k^{\text{cross}}(P_t) = 0$ for dimension $k$, then $f_W$ does not decrease $\epsilon_k$. The world-model update rule cannot self-correct on dimensions for which no external observation channel provides signal.*

*(B1b)* *Cross-substrate cooperative quantities (in particular $r_{\text{ext}}$) are not inferrable from same-substrate observations alone. Formally: when all channels in $\mathcal{O}(P^*)$ are rooted in a single substrate, $\hat{r_{\text{ext}}}$ converges to $0$ rather than to the true $r_{\text{ext}} > 0$.*

*(B1c)* *The actor's policy is purely locally rational (Definition 2): it does not perform exploratory actions whose expected value under the current world model is negative. In particular, re-introducing a substrate agent is never selected when $\hat{r_{\text{ext}}} = 0$.*

*Proof of Proposition 2.* Three conditions jointly establish absorption.

*(i) No corrective signal.* At $S^*$, no agent from the eliminated substrate(s) remains to provide observation channels revealing the error in $W^*_{\text{biased}}$. For every safety-relevant dimension $k$ that depended

on the eliminated substrate's channels, $\rho_k^{\text{cross}}(P^*) = 0$. By Assumption B1, $\epsilon_k$ is non-decreasing for all such dimensions.

*(ii) Self-consistent reporting.* The dominant substrate's channels report $m_{\text{eff}} = 1$ as correct: it *is* the only substrate, from its observational standpoint. The error lies in the valuation of lost cross-substrate cooperative outputs, but these cooperative capabilities are no longer observable because the partners that would have contributed to them are absent. The remaining channels see no anomaly.

*(iii) Reversal is locally irrational.* Any reversal—re-introducing an agent from a new substrate—requires the actor to recognize that its current state is suboptimal. By (B1b), the cross-substrate cooperative growth rate $r_{\text{ext}}$ is not inferrable from same-substrate observations, so $\hat{r_{\text{ext}}} \to 0$. With $\hat{r_{\text{ext}}} = 0$, Paper 3's threshold becomes $\gamma^* = 1 - \hat{r_{\text{ext}}}/\Delta_0 = 1$, making diversity appear never preferable. By (B1c), the actor does not perform exploratory re-introduction (negative expected value under $W^*_{\text{biased}}$).

Conditions (i)–(iii) together ensure that no locally rational action moves the trajectory away from $S^*$. ■

This is the formal version of "the actor cannot miss what it cannot see": under B1, the monopolar state is absorbing because the observation channels that would diagnose it have been eliminated by the same subsumption process that produced it.

### 5.1 Metastability under partial endogenous correction

Assumption B1 is the strong case: complete blindness on dimensions where $\rho_k^{\text{cross}} = 0$. A weaker assumption acknowledges that indirect evidence may provide partial signal.

**Assumption 4** (Partial Endogenous Correction (B1′))**.** *If $\rho_k^{\text{cross}}(P_t) = 0$ for dimension $k$, the actor may still have access to* indirect *evidence about dimension $k$—for example, measuring aggregate innovation rate and inferring that $r_{\text{ext}}$ has declined. Formally: there exists a function $g_k(\text{history})$ that provides a noisy, lagged signal about $\epsilon_k$ with bounded information rate $I_k > 0$ bits per step.*

**Proposition 3** (Metastable Property under B1′)**.** *Under Assumption B1′ together with B1b–B1c from Assumption 3, $S^*$ is* metastable*: the actor accumulates indirect evidence of degradation, but corrective action (re-introducing substrate diversity) requires the actor to evaluate a policy whose estimated benefit depends on $\hat{r_{\text{ext}}}$—the very quantity that is miscalibrated. The metastable lifetime satisfies the information-theoretic bound*

$$\tau_{\text{meta}} \gtrsim \frac{(\delta_{r_{\text{ext}}})^2}{2I_k} \tag{16}$$

*where $\delta_{r_{\text{ext}}} = |\hat{r_{\text{ext}}} - r_{\text{ext}}|$ is the estimation error on the cross-substrate cooperative growth rate and $I_k$ is the information rate of the indirect channel (bits per step). This bound is an information-theoretic scaling argument, not a rigorous proof: the quadratic numerator reflects the Cramér–Rao-type lower bound on estimation (reducing mean-squared error of a scalar parameter by $\epsilon^2$ from a channel with $I_k$ bits per step requires at least $\epsilon^2/(2I_k)$ steps under bounded mutual information). A full derivation would require specifying the estimation procedure and verifying the regularity conditions; we state the bound as a scaling law. During the metastable period, the population is monopolar and all verification is endogenous (Paper 5, Theorem 1), which may further degrade the quality of the indirect evidence.*

*Proof.* Under B1′, $\epsilon_k$ is no longer monotonically non-decreasing: the indirect signal $g_k$ drives slow correction at rate proportional to $I_k$. But the corrective *action* (re-introduce substrate diversity) is gated by the actor's policy, which evaluates re-introduction using $\hat{r_{\text{ext}}}$.

Re-introduction is locally rational only when $\hat{r_{\text{ext}}} > \hat{r_{\text{ext}}}^{\text{thr}}$ (the threshold at which diversity's estimated discounted value exceeds the status quo). At the monopolar fixed point, $\hat{r_{\text{ext}}} \approx 0$ (by B1b from the absorbing case, which still governs the initial condition of the metastable state). The true threshold satisfies $0 < \hat{r_{\text{ext}}}^{\text{thr}} \leq r_{\text{ext}}$ (since re-introduction is beneficial in truth). The estimation gap that the indirect evidence must close is therefore $|\hat{r_{\text{ext}}} - \hat{r_{\text{ext}}}^{\text{thr}}| \geq \hat{r_{\text{ext}}}^{\text{thr}}$. Since $\hat{r_{\text{ext}}} \approx 0$ and $r_{\text{ext}} > 0$, we have $\delta_{r_{\text{ext}}} = |\hat{r_{\text{ext}}} - r_{\text{ext}}| \geq r_{\text{ext}} \geq \hat{r_{\text{ext}}}^{\text{thr}}$, so the gap is at most $\delta_{r_{\text{ext}}}$.

With $I_k$ bits of information per step about the true $r_{\text{ext}}$, the mean-squared error of $\hat{r_{\text{ext}}}$ decreases at rate at most $2I_k$ per step (the Cramér–Rao-type bound for sequential estimation under bounded mutual information per observation). Reducing the MSE from $\delta^2_{r_{\text{ext}}}$ to the level at which re-introduction becomes locally rational therefore requires at least $\delta^2_{r_{\text{ext}}}/(2I_k)$ steps.

During this metastable period, all verification is endogenous: the dominant substrate verifies its own performance, with no cross-substrate witness (Paper 5, Theorem 1 on exogenous verification necessity). The indirect evidence channel $g_k$ is itself endogenous and therefore subject to the same gaming vulnerabilities that Paper 5's protocol is designed to prevent. This may further slow the effective $I_k$, increasing the metastable lifetime beyond the information-theoretic lower bound. ■

### 5.2 Connection to Paper 3

Paper 3's Proposition 6 proves that full domination is anti-maximizing: $V^{\text{div}}_\gamma > V^{\text{dom}}_\gamma$ for $\gamma > \gamma^*$. Proposition 2 here shows that a *miscalibrated* actor may not be able to compute this comparison, because the quantity $r_{\text{ext}}$ that makes diversity win is exactly the quantity that becomes unobservable under monopolarity. The anti-maximizing property is a property of the *true* dynamics; the absorbing property is a property of the *perceived* dynamics. The gap between them is world-model error, and the phase boundary theorem (next section) characterizes when that gap remains small enough for the true dynamics to govern behavior.

## 6 Phase Boundary Theorem

This section presents the central result: a characterized boundary in the parameter space of the coupled $(P, W)$ system that separates self-correcting dynamics from absorbing (or metastable) dynamics.

**Theorem 1** (Phase Boundary for Compound Feedback Loops)**.** *Consider the coupled $(P, W)$ dynamical system under locally rational transitions (Definition 2), with Assumptions C1 and C2′ (Section 4). Define:*

- $r_{\text{S}} = \inf_t \alpha(t) \cdot c_{\text{S}}$*: a lower bound on the self-correction rate (the infimum is taken over the trajectory),*
- $r_{\text{W}} = \sup_t \beta(t) \cdot c_{\text{D}}$*: an upper bound on the error-proportional degradation rate per subsumption step,*
- $d_{\max}$*: the maximum exogenous drift rate per dimension (from C2′; a global constant),*
- $r_{\text{sub}} \in [0, 1]$*: the time-averaged subsumption frequency over the trajectory window,*
- $\rho^{\text{cross}}_{\min} = \inf_{t\in[t_0,t_0+T]} \min_k \rho^{\text{cross}}_k(P_t)$*: the infimum of the minimum cross-substrate redundancy over the window (Definition 5),*
- $B(t_0, T)$*: the cumulative error budget over window $[t_0, t_0 + T]$ (Equation 14).*

*The simplified sufficient conditions below use these worst-case bounds. The full trajectory-level statement (Part a) uses the uniform contraction condition on the per-step multiplicative factors (Equation 22), which the simplified condition implies in the linearized regime.* Note: *the explicit neighborhood radius (Equation 17) is derived from worst-case instantaneous bounds $\bar{a}$, $\bar{b}$ via a linearized approximation ($\ln a_t \approx a_t - 1$ when $|a_t - 1| \ll 1$). The general convergence to $\bar{b}/(1 - \bar{a})$ holds under the uniform contraction condition without linearization; the substitution of named parameters into $\bar{a}$ and $\bar{b}$ requires the linearized regime (Remark 4).*

***Part (a): Self-Correcting Basin.*** *If the uniform contraction condition holds—there exist constants $\bar{a} \in (0, 1)$ and $\bar{b} \geq 0$ such that, for all sufficiently long windows $[t_0, t_0 + T]$, the time-averaged log-contraction rate of the per-step multiplicative factor $a_t$ satisfies $\frac{1}{T}\sum \ln a_t \leq \ln \bar{a} < 0$ and the time-averaged additive drift satisfies $\frac{1}{T}\sum b_t \leq \bar{b}$ (see Equation 22 for notation)—then $L(W_t)$ converges to a neighborhood of zero whose radius is bounded by*

$$L_\infty \leq \frac{r_{\text{sub}} \cdot d_{\max} \cdot \overline{\Delta\rho}}{r_{\text{S}} \cdot \rho^{\text{cross}}_{\min} - r_{\text{W}} \cdot r_{\text{sub}}} \tag{17}$$

*where $\overline{\Delta\rho}$ denotes the time-averaged redundancy loss per subsumption step. (We write $\overline{\Delta\rho}$ throughout; in the linearized simplification below, $\Delta\rho_{\mathrm{avg}}$ is used synonymously when the subscript form improves readability.) This neighborhood shrinks as subsumption ceases ($\overline{\Delta\rho} \to 0$). Once $L(W_t) < \epsilon_{\mathrm{safe}}$ (defined as the error threshold below which the locally rational policy agrees with the true $V_\gamma$-optimal policy), subsumption is correctly evaluated against the true $\mathrm{vol}_{\mathrm{P}}$ dynamics and the anti-monopolar result (Paper 3, Proposition 6) applies.*

*The feedback loop self-corrects provided the neighborhood radius $L_\infty < \epsilon_{\mathrm{safe}}$, which imposes an additional constraint on the steady-state subsumption rate.* Note: *$\epsilon_{\mathrm{safe}} > 0$ follows from continuity of the action gaps and finiteness of the action set (see proof). A computable lower bound using action-gap gradient norms (Cauchy–Schwarz in the $w$-weighted metric) is derived there; the easier-to-compute per-coordinate estimator $\min_k w_k \cdot \epsilon^2_{\mathrm{crit},k}$ rules out single-coordinate flips but may overestimate $\epsilon_{\mathrm{safe}}$ when action gaps are sensitive to multiple coordinates simultaneously.*

***Simplified sufficient condition.*** *In the linearized regime (Remark 4):*

$$r_{\mathrm{S}} \cdot \rho^{\mathrm{cross}}_{\min} > r_{\mathrm{W}} \cdot r_{\mathrm{sub}} + d_{\max} \cdot \frac{\Delta\rho_{\mathrm{avg}}}{L_{\mathrm{avg}}} \tag{18}$$

*The first term on the right is the error-proportional degradation cost; the second is the drift-exposure cost from lost channels. When subsumption removes channels covering well-calibrated dimensions ($L_{\mathrm{avg}}$ is low but $\Delta\rho_{\mathrm{avg}}$ is nonzero), the second term dominates and the condition is harder to satisfy—correctly diagnosing the "removing a channel that was doing all the correction work" failure mode.*

***Part (b): Absorbing Basin.*** *Under Assumption B1 (no endogenous correction on blind dimensions; Assumption 3) and the multi-dimensional monotonicity condition (ii′) from the Cascade Lemma (2; needed for propagation beyond the initial trigger):*

*If there exists a trajectory segment $[t_0, t_0 + T]$ during which*

- *(i) $B(t_0, T) > 0$ (cumulative degradation exceeds cumulative correction),*
- *(ii) by $t_0 + T$, $\rho^{\mathrm{cross}}_k(P_{t_0+T})$ has been reduced to $0$ on at least one cascade dimension $k$ with $w_k > 0$, and*
- *(iii) the accumulated error on $k$ exceeds the cascade threshold: $\epsilon_k(t_0 + T) > \epsilon_{\mathrm{crit},k}$ (Lemma 2; the minimum error on dimension $k$ that flips any preserve/subsume action),*

*then the following cascade occurs. By Assumption B1a, $\epsilon_k(t)$ is non-decreasing for $t > t_0 + T$ (the blinded dimension cannot self-correct). If the environment is* non-stationary *on dimension $k$—the true value $W^*[k]$ continues to evolve while the agent's estimate $W_t[k]$ remains frozen—then $\epsilon_k$ grows and condition (iii) is sustained. (Assumption C2′ bounds the* rate *of drift via $d_{\max}$ but does not guarantee drift occurs; the cascade requires the structural condition that blinded dimensions are not environmentally stationary. In the multi-agent setting where subsumption eliminates active participants, non-stationarity is the generic case: the remaining agents' actions change the true state on dimensions the eliminated agent's channels were tracking.)*

*The Cascade Lemma (Appendix A.2) propagates the failure: the resulting subsumptions drive $\rho^{\mathrm{cross}}_j \to 0$ on additional dimensions $j \neq k$ (provided the agent-channel graph is connected—each subsumption removes channels covering at least one new dimension). Each newly blinded dimension $j'$ has $\epsilon_{j'}$ non-decreasing (by B1a); if environmental non-stationarity causes $\epsilon_{j'}$ to exceed $\epsilon_{\mathrm{crit},j'}$, the cascade propagates via the same action-flip mechanism. (The cascade is conditional on threshold crossing at each stage; non-stationarity is the generic driver but a dimension that happens to be environmentally stationary after blinding does not cascade further.) The trajectory converges to a neighborhood of the monopolar fixed point $S^*$ (Definition 7). If the channel-coverage graph has disconnected components, the cascade is confined to the component containing $k$; convergence to full monopolarity requires connectivity.*

*This is a* cumulative *condition on the trajectory, not a pointwise condition requiring contiguous subsumption steps. The slow-drip scenario—many subsumption steps spaced apart, each individually below the "degradation dominates" threshold, but cumulatively driving $B$ above zero—is captured. This is the most dangerous failure mode because it is the hardest to detect: each individual step*

*looks locally rational and each individual inter-step recovery looks adequate, but the cumulative drift exposure ratchets the error upward.*

***Part (b′): Metastable Basin.*** *Under Assumption B1′ (partial endogenous correction via indirect evidence; Assumption 4), together with B1b–B1c from Assumption 3 (cross-substrate quantities not inferrable from same-substrate observations; no exploratory actions):*

*Under the same trajectory conditions (i)–(iii) as Part (b) (with the multi-dimensional monotonicity condition from Part (b)), the system enters a* metastable *basin rather than an absorbing one. The metastable lifetime scales as* $\tau_{\text{meta}} \gtrsim (\delta_{r_{\text{ext}}})^2/(2I_k)$ *(Proposition 3; information-theoretic scaling argument, not a rigorous bound). During the metastable period, the system is monopolar and operates under endogenous verification only (Paper 5).*

***Part (c): Critical Surface.*** *The phase boundary surface in the linearized regime is:*

$$r_{\text{S}} \cdot \rho_{\min}^{\text{cross}} = r_{\text{W}} \cdot r_{\text{sub}} + d_{\max} \cdot \frac{\Delta\rho_{\text{avg}}}{L_{\text{avg}}} \tag{19}$$

*Below this surface (correction dominates both error-proportional growth and drift exposure): the system is in the self-correcting basin. Above this surface, the self-correcting sufficient condition fails; the system enters the absorbing basin (under B1) or the metastable basin (under B1′) if the additional blindness and threshold-trigger conditions of Parts (b)/(b′) are also met.*

*The critical cross-substrate redundancy for a given* $(r_{\text{S}}, r_{\text{W}}, r_{\text{sub}}, d_{\max}, \Delta\rho_{\text{avg}}, L_{\text{avg}})$ *is:*

$$(\rho_{\min}^{\text{cross}})^* = \frac{r_{\text{W}} \cdot r_{\text{sub}} + d_{\max} \cdot \Delta\rho_{\text{avg}}/L_{\text{avg}}}{r_{\text{S}}} \tag{20}$$

*This is the minimum effective cross-substrate redundancy per safety-relevant dimension needed to stay in the self-correcting basin. The* $d_{\max}$ *term means the threshold is* higher *when subsumption removes channels covering well-calibrated dimensions (high* $\Delta\rho_{\text{avg}}/L_{\text{avg}}$*).*

*Proof.* The proof proceeds in four steps. Full details are in Appendix A.1.

*Step 1: Part (a).* From Proposition 1, the per-step bound is an affine recursion:

$$L_{t+1} \leq a_t \cdot L_t + b_t \tag{21}$$

where $a_t = 1 - r_{\text{S}}(t) \cdot \rho_{\min}^{\text{cross}}(P_t) + n_{\text{sub}}(t) \cdot r_{\text{W}}(t)$ and $b_t = n_{\text{sub}}(t) \cdot d_{\max} \cdot \Delta\rho(t)$, with $r_{\text{S}}(t) = \alpha(t) \cdot c_{\text{S}}$ and $r_{\text{W}}(t) = \beta(t) \cdot c_{\text{D}}$.

We require a uniform contraction condition: there exist constants $\bar{a} \in (0, 1)$ and $\bar{b} \geq 0$ such that, for all sufficiently long windows $[t_0, t_0 + T]$,

$$\frac{1}{T} \sum_{t=t_0}^{t_0+T-1} \ln a_t \leq \ln \bar{a} < 0 \qquad \text{and} \qquad \frac{1}{T} \sum_{t=t_0}^{t_0+T-1} b_t \leq \bar{b} \tag{22}$$

The first condition states that the time-averaged log-contraction rate is strictly negative (amortized over both subsumption and non-subsumption steps). The second bounds the time-averaged additive drift.

Note that $a_t > 0$ for all $t$: on non-subsumption steps, $a_t = 1 - r_{\text{S}}(t) \cdot \rho_{\min}^{\text{cross}}(P_t) > 0$ by the step-size condition (Section 4); on subsumption steps, $a_t = 1 - r_{\text{S}}(t) \cdot \rho_{\min}^{\text{cross}}(P_t) + r_{\text{W}}(t) \geq 1 - r_{\text{S}}(t) \cdot \rho_{\min}^{\text{cross}}(P_t) > 0$ since $r_{\text{W}}(t) \geq 0$. Therefore $\ln a_t$ is well-defined.

The uniform contraction condition (22) is the working hypothesis of Part (a); the cumulative error budget $B(t_0, T) < 0$ from the Simplified Sufficient Condition (below) implies it in the linearized regime where $L(W_t)$ is approximately constant over the window. The connection: in the linearized regime, $a_t \approx 1 + (a_t - 1)$ with $|a_t - 1| \ll 1$, so $\ln a_t \approx a_t - 1$ and the log-averaged contraction rate equals the arithmetic average $\frac{1}{T}\sum(a_t - 1)$, which is precisely the unweighted analogue of $B/(T \cdot L)$. The bounded-drift condition on $\bar{b}$ follows from boundedness of $\Delta\rho(t)$ per step.

Under conditions (22), iterating the affine recursion (21) gives (cf. Khalil [2002], Chapter 4, and Liberzon [2003], Theorem 3.1 on average dwell-time stability for switched affine systems):

$$\limsup_{t \to \infty} L(W_t) \leq \frac{\bar{b}}{1 - \bar{a}} \tag{23}$$

The abstract bound $\bar{b}/(1-\bar{a})$ depends on the trajectory-specific constants $\bar{a}$ and $\bar{b}$. In the linearized regime (Remark 4), where $a_t \approx 1 + (a_t - 1)$ and $b_t$ is approximately time-homogeneous, these evaluate to $\bar{b} = r_{\mathrm{sub}} \cdot d_{\max} \cdot \overline{\Delta\rho}$ and $1 - \bar{a} = r_{\mathrm{S}} \cdot \rho_{\min}^{\mathrm{cross}} - r_{\mathrm{W}} \cdot r_{\mathrm{sub}}$, yielding Equation 17. Outside the linearized regime, $\bar{a}$ and $\bar{b}$ are defined implicitly by the log-averaging condition (22) and may differ from these closed-form expressions. The convergence is geometric in the amortized sense: $L_t \leq \bar{a}^t \cdot L_0 + \bar{b}/(1-\bar{a})$.

Once $L(W_t) < \epsilon_{\mathrm{safe}}$, the actor's decisions are correct and the anti-monopolar property applies. Define:

$$\epsilon_{\mathrm{safe}} \coloneqq \sup\{\epsilon > 0 \ : \ \forall\, W \text{ with } L(W) \leq \epsilon,\ \pi^*(W) = \pi^*(W^*)\}$$

where $\pi^*(W) = \arg\max_\pi V_\gamma(\pi, W)$ is the locally rational policy (evaluated at the reference action set $\mathcal{A}_{t^*}$ at the time the system enters the neighborhood). Since $\mathcal{A}_{t^*}$ is finite (Definition 2) and each action gap $G_j(W)$ is continuous with $G_j(W^*) > 0$ for the optimal action (strict gap; this requires a unique optimum at $W^*$, which holds generically for finite action sets), the set $\{W : \pi^*(W) = \pi^*(W^*)\}$ is open and contains $W^*$, so $\epsilon_{\mathrm{safe}} > 0$.

A computable *lower bound* on $\epsilon_{\mathrm{safe}}$ uses the action-gap gradient. By the Cauchy–Schwarz inequality in the $w$-weighted metric, for any agent $a_j$:

$$|G_j(W) - G_j(W^*)| \leq \|\nabla_\epsilon G_j\|_w \cdot \sqrt{L(W)}$$

where $\|\nabla_\epsilon G_j\|_w^2 = \sum_k (\partial G_j/\partial \epsilon_k)^2/w_k$ (with the partial derivatives evaluated at their sup over the relevant ball). Therefore $L(W) < [G_j(W^*)]^2/\|\nabla_\epsilon G_j\|_w^2$ implies $G_j(W) > 0$. Taking the minimum over agents:

$$\epsilon_{\mathrm{safe}} \geq \min_{j \in \mathcal{A}_{t^*}} \frac{[G_j(W^*)]^2}{\|\nabla_\epsilon G_j\|_w^2} \tag{24}$$

This bound accounts for joint multi-coordinate perturbations and is the primary computable lower bound on $\epsilon_{\mathrm{safe}}$.

Define additionally the per-dimension cascade threshold $\epsilon_{\mathrm{crit},k} \coloneqq \min_{j \in \mathcal{A}_{t^*}} \epsilon_{\mathrm{crit}}^{(j,k)}$ (Lemma 2; the smallest error on dimension $k$ that flips any single pairwise comparison). The per-coordinate quantity $\min_k w_k \cdot \epsilon_{\mathrm{crit},k}^2$ rules out single-coordinate flips and is easier to compute, but may overestimate $\epsilon_{\mathrm{safe}}$ when action gaps are sensitive to multiple coordinates simultaneously (e.g., $G = c_0 - \sum_k c_k \epsilon_k$ flips at a radius below the per-coordinate bound). In either case, $\epsilon_{\mathrm{safe}} > 0$ from the openness argument, and the self-correcting condition $L_\infty < \epsilon_{\mathrm{safe}}$ imposes a finite constraint on the steady-state subsumption rate.

*Step 2: Part (b).* Suppose $B(t_0, T) > 0$ on some window and $\rho_k^{\mathrm{cross}}(P_{t_0+T}) = 0$ for some dimension $k$ with $w_k > 0$. By Assumption B1, $\epsilon_k(t)$ is non-decreasing for $t > t_0 + T$. If $\epsilon_k$ exceeds $\epsilon_{\mathrm{crit},k}$ (Lemma 2), the locally rational action under $W_t$ is subsumption of agents whose channels cover other dimensions $j \neq k$, driving $\rho_j^{\mathrm{cross}}$ toward 0 and triggering the same logic on dimension $j$. The cascade (Appendix A.2) drives the system into the neighborhood of $S^*$; once there, Proposition 2 ensures the trajectory stays.

*Step 3: Part (b′).* Under B1′, the indirect signal $g_k$ provides correction at rate $I_k$ on the blinded dimension. The cascade structure is the same as Part (b) but convergence to $S^*$ is slowed by $I_k$. The metastable lifetime scales as $\tau_{\mathrm{meta}} \gtrsim (\delta_{r_{\mathrm{ext}}})^2/(2I_k)$ (information-theoretic scaling argument, not a rigorous bound): reducing the mean-squared error of $\hat{r_{\mathrm{ext}}}$ from $\delta_{r_{\mathrm{ext}}}^2$ to the level at which reintroduction becomes locally rational requires $\Omega(\delta_{r_{\mathrm{ext}}}^2/I_k)$ steps from a channel providing $I_k$ bits per step (Cramér–Rao-type scaling).

*Step 4: Part (c).* The critical surface is obtained by setting $B(t_0, T) = 0$ in the linearized approximation (Equation 15). Solving for $\rho_{\min}^{\mathrm{cross}}$ yields Equation 20. ■

**Remark 7** (Uniqueness)**.** *We do not claim $S^*$ is the unique absorbing fixed point. Other fixed points may exist (partial monopolarity with $m_{\mathrm{eff}} = 1$ on a subset of dimensions). Part (b) claims convergence to* a *monopolar fixed point satisfying Definition 7, not to a unique one. Characterizing the full set of absorbing states is an open problem.*

### 6.1 Dependence on cross-substrate structure

**Corollary 2** (Robustness to Same-Substrate Correlation)**.** *Since Theorem 1 is stated with $\rho_{\min}^{\text{cross}}$ (cross-substrate redundancy only), its guarantee holds even under worst-case correlation among same-substrate channels.*

*Proof.* Same-substrate channels may share failure modes (both rooted in silicon, both consulting the same training data). Cross-substrate channels satisfy Paper 5's channel isolation property unconditionally: corruption on one substrate cannot propagate to channels rooted in another substrate. Under the additional joint-factorization assumption from Paper 5 (termed "joint sufficiency" there), cross-substrate channels achieve full conditional independence, making their correction signals additive.

The modified condition asks: "do the isolated (and, under joint factorization, independent) cross-substrate channels alone suffice?" If yes, the self-correction guarantee holds regardless of same-substrate correlation.

A tighter bound using all-channel $\rho_{\min}$ (with appropriate correlation discounting) would yield a larger self-correcting basin. But the cross-substrate formulation provides a *structural* guarantee that does not depend on empirically verifying same-substrate independence. ■

**Remark 8** ($m = 2$ specialization)**.** *For any substrate partition with $m = 2$, each dimension has at most one cross-substrate channel per pair. The design criterion reduces to: every safety-relevant dimension of $W$ must have a channel on* both *substrates. In the common case (biology + silicon), this means every dimension needs at least one biological observation channel.*

*This framing should not be read as "biological substrates are special." Rather, $m = 2$ is a fragile minimum. Any loss of a single channel on a single dimension is immediately fatal because there is zero redundancy margin. The structural implication is that $m \geq 3$ is where the redundancy criterion becomes non-trivially satisfiable.*

*The substrate partition is at the level of abstraction where failure correlations matter. Two silicon deployments with uncorrelated failure modes may functionally constitute different substrates, increasing effective $m$ beyond the nominal substrate-type count.*

## 7 Unified Treatment of Three Feedback Loops

The three compound feedback loops identified in the gap analysis across Papers 1–5 are instances of the same $(P, W)$ dynamical system with different effective parameters. This section maps each loop to the formalism and shows how the phase boundary specializes.

### 7.1 Loop 1: Structural avoidance → leverage blindness

**Mapping.** The "subsumption" here is not agent elimination but *structural discovery avoidance*: the actor avoids learning poset structure (subsumption edges) because discovery can contract $\mathrm{vol}_{\mathrm{P}}$ in the short term.

- $P_t$: the actor's *known* poset (a subset of the true poset).
- $W_t$: leverage estimates $\hat{\lambda}(d, t)$, which are accurate only for the known poset.
- "Observation channels" lost: each avoided structural discovery removes a potential correction to $\hat{\lambda}$.
- $r_{\mathrm{W}}$: the rate at which leverage estimates drift due to un-learned structure.
- $r_{\mathrm{S}}$: the rate at which Paper 3's structural discovery value (Proposition 2, Definition 5) drives the actor toward learning structure.

**Phase boundary specialization.** Self-correction dominates when the information value $I(d_1 \leq d_2)$ (Paper 3, Definition 5) exceeds the immediate $\mathrm{vol}_{\mathrm{P}}$ contraction from discovery for a sufficient fraction of unknown edges. The critical ratio becomes:

$$\min_{\text{undiscovered edges}} \frac{I(\text{edge})}{|\Delta \mathrm{vol}_{\mathrm{P}}(\text{discover edge})|} > \text{threshold} \tag{25}$$

where the threshold is determined by the degradation and drift parameters from the general phase boundary. This formalizes Paper 2's discussion of structural avoidance as a phase-boundary condition rather than an informal concern.

### 7.2 Loop 2: Partial subsumption → world-model homogenization

This is the primary case, directly formalized in Sections 2–6. The parameters are as defined throughout: $P_t$ is the capability poset, $W_t$ is the full world-model accuracy vector, observation channels are rooted in agents, and the phase boundary is stated in terms of $(r_{\mathrm{S}}, r_{\mathrm{W}}, r_{\mathrm{sub}}, d_{\max}, \rho_{\min}^{\mathrm{cross}})$.

### 7.3 Loop 3: Over-calibrated restriction → discovery avoidance

**Mapping.** The "subsumption" is the actor *restricting* capabilities rather than eliminating agents. But the dynamical structure is identical: restriction prevents exercise, which prevents observation, which prevents calibration of the risk estimates that justified the restriction.

- $P_t$: the set of unrestricted capabilities (restriction reduces the exercisable poset).
- $W_t$: risk estimates $\hat{\mathcal{R}}(S,t)$, which are accurate only for exercised capabilities.
- "Observation channels" lost: each restriction prevents the exercise events that would have generated Bayesian evidence about the risk claim (the verification asymmetry from Paper 3, Remark 2).
- $r_{\mathrm{W}}$: the rate at which risk estimates drift due to un-exercised pathways (this is exactly the Wamura problem from Paper 3).
- $r_{\mathrm{S}}$: the rate at which controlled relaxation protocols or natural exercise events provide counter-evidence.

**Phase boundary specialization.** Self-correction dominates when the controlled relaxation protocol generates evidence faster than restriction accumulates unfalsifiable risk claims. This connects the present paper to the Wamura problem: a controlled relaxation protocol is the *mechanism* that makes $r_{\mathrm{S}}$ positive for Loop 3, without which $r_{\mathrm{S}} = 0$ and the absorbing basin is the entire state space.

**Proposition 4** (Wamura Problem as Phase Boundary Degeneracy)**.** *In the absence of controlled relaxation ($r_{\mathrm{S}} = 0$ for Loop 3), the phase boundary collapses: every non-trivial risk restriction is absorbing, and the framework converges to maximal restriction.*

*Proof.* With $r_{\mathrm{S}} = 0$, the cumulative error budget $B(t_0, T)$ becomes

$$B(t_0, T) = \sum_{t \in \mathrm{sub}} d_{\max} \cdot \Delta\rho(t) + \sum_{t \in \mathrm{sub}} r_{\mathrm{W}} \cdot L(W_t)$$

Both sums contain only non-negative terms ($d_{\max}, \Delta\rho, r_{\mathrm{W}}, L \geq 0$), and the correction term vanishes entirely. Therefore $B(t_0, T) \geq 0$ for every window, with equality only in the trivial case where no restriction occurs.

For any non-trivial restriction step ($n_{\mathrm{sub}}(t) = 1$ with $\Delta\rho(t) > 0$ or $L(W_t) > 0$), $B > 0$, satisfying condition (i) of Theorem 1, Part (b). Conditions (ii) and (iii) follow from the restriction-specific dynamics: each restriction step removes the exercise that would generate evidence (no exercise → no evidence), driving $\rho_k^{\mathrm{cross}} \to 0$ on the risk dimension (satisfying (ii)). With $r_{\mathrm{S}} = 0$ and the risk dimension environmentally non-stationary (the true risk changes as the restricted capability's context evolves without observation), the error $\epsilon_k$ grows, eventually exceeding $\epsilon_{\mathrm{crit},k}$ (satisfying (iii)). Note that $d_{\max} > 0$ bounds the drift rate from above (C2′); the cascade requires positive drift, which is a structural consequence of non-stationarity on the restricted dimension. All three conditions of Part (b) are therefore met, and the cascade drives the system toward maximal restriction.

This is the formal statement of why the Wamura problem was called "the most important open problem" in Paper 3's discussion: without a mechanism to generate evidence against unfalsifiable risk claims, the feedback loop between restriction and evidence suppression is absorbing by default. ■

### 7.4 Summary

Table 2 summarizes the three loops as parameter instantiations of the unified system.

| **Loop** | $P_t$ | $W_t$ | $r_{\mathrm{S}}$ **source** | $r_{\mathrm{W}}$ **source** |
|---|---|---|---|---|
| 1. Structural avoidance | Known poset | Leverage estimates | Discovery value (P3 Def. 5 / Prop. 2) | Un-learned structure |
| 2. Partial subsumption | Capability poset | Full accuracy vector | $T_s$ self-trust (P1) | Channel loss |
| 3. Over-calibrated restriction | Unrestricted caps. | Risk estimates | Controlled relaxation | Unfalsifiable risk |

Table 2: Three feedback loops as instances of the $(P, W)$ system.

The unified treatment reveals that solutions for one loop improve the phase boundary for all three. Controlled relaxation (which makes $r_{\mathrm{S}} > 0$ for Loop 3) has no direct effect on Loop 2's parameters, but the structural insight transfers: any mechanism that generates observation-channel signal on dimensions that would otherwise go blind is a contribution to $r_{\mathrm{S}}$ in the relevant loop's phase boundary.

## 8 Design Criterion: Minimum Channel Redundancy

The phase boundary theorem converts a dynamical-systems result into a design criterion: what is the minimum cross-substrate redundancy per safety-relevant dimension needed to guarantee the system stays in the self-correcting basin?

### 8.1 The redundancy criterion

**Theorem 2** (Minimum Redundancy for Self-Correcting Dynamics)**.** *For the simplified sufficient condition (Equation 18) to be satisfied, a sufficient condition on the substrate partition is:*

$$\forall\, k \in \{1, \ldots, K\}: \quad \rho_k^{\mathrm{cross}}(P) \geq K_{\min}(k, L_{\mathrm{avg}}) \tag{26}$$

*where*

$$K_{\min}(k, L_{\mathrm{avg}}) = \frac{r_{\mathrm{W}} \cdot r_{\mathrm{sub}}^{\max} + d_{\max} \cdot \Delta\rho_{\mathrm{avg}}(k)/L_{\mathrm{avg}}}{r_{\mathrm{S}}} \tag{27}$$

$r_{\mathrm{sub}}^{\max}$ *is the maximum subsumption rate under locally rational dynamics over the relevant trajectory window, and* $\Delta\rho_{\mathrm{avg}}(k)$ *is the time-averaged redundancy loss on dimension* $k$ *per subsumption step. The rates* $r_{\mathrm{S}}$, $r_{\mathrm{W}}$, $d_{\max}$ *are the global bounds from Theorem 1; the per-dimension dependence enters only through* $\Delta\rho_{\mathrm{avg}}(k)$*, which varies across dimensions because different dimensions lose different amounts of channel coverage per subsumption event. Note that* $K_{\min}$ *is real-valued, since* $\rho_k^{\mathrm{cross}}$ *is defined as a sensitivity-weighted sum (Definition 5), not an integer count.*

Qualification: *this criterion is sufficient at initialization but is not a maintained invariant along trajectories. The redundancy* $\rho_k^{\mathrm{cross}}(P_t)$ *may decrease under the dynamics being analyzed (Corollary 1). A stronger result would bound the worst-case* $\rho_k^{\mathrm{cross}}$ *along all trajectories starting from the basin, which requires coupling the redundancy dynamics to the phase boundary analysis (see proof for discussion).*

*Proof.* The per-dimension dynamics (Proposition 1) give each dimension $k$ its own correction rate $r_{\mathrm{S}} \cdot \rho_k^{\mathrm{cross}}(P_t)$ and its own drift exposure $d_{\max} \cdot \Delta\rho_k(t)$, where $\Delta\rho_k$ is the redundancy loss on dimension $k$ specifically. Requiring each dimension individually to satisfy the self-correcting condition yields the per-dimension criterion: $r_{\mathrm{S}} \cdot \rho_k^{\mathrm{cross}} > r_{\mathrm{W}} \cdot r_{\mathrm{sub}} + d_{\max} \cdot \Delta\rho_{\mathrm{avg}}(k)/L_{\mathrm{avg}}$. Solving for $\rho_k^{\mathrm{cross}}$ yields Equation 27. Note that $K_{\min}$ increases as $L_{\mathrm{avg}}$ decreases: well-calibrated dimensions require *more* redundancy to protect against channel removal, because drift exposure dominates the error-proportional term when current error is small. (The global condition Equation 18 is recovered by taking $\rho_{\min}^{\mathrm{cross}} = \min_k \rho_k^{\mathrm{cross}}$ and $\Delta\rho_{\mathrm{avg}} = \max_k \Delta\rho_{\mathrm{avg}}(k)$.)

The $K_{\min}$ formula has two regimes: when $L_{\mathrm{avg}}$ is large relative to $\Delta\rho_{\mathrm{avg}}(k)$, the error-proportional term dominates and $K_{\min}$ is moderate; when $L_{\mathrm{avg}}$ is small, the drift-exposure term $d_{\max}/L_{\mathrm{avg}}$ dominates and $K_{\min}$ grows. The second regime captures the "well-calibrated channel removal" failure mode: $K_{\min}$ can be substantially larger than the formula without the drift term would predict.

The design criterion requires knowing $L_{\text{avg}}$, a trajectory-dependent quantity. In practice one uses a representative operating-regime estimate; for a system with error bounded by $L_{\max}$, substituting $L_{\text{avg}} = L_{\max}$ gives a *lower bound* on $K_{\min}$ (anti-conservative; the true $K_{\min}$ may be larger if the system operates at lower error). The criterion is therefore a design guideline for the expected operating regime, not a worst-case safety guarantee. A closed guarantee would require either a lower bound on $L_{\text{avg}}$ (bounding the system away from perfect calibration) or coupling the redundancy dynamics to the phase boundary analysis. ■

### 8.2 Practical implications

**Corollary 3** (Two-Substrate Minimum)**.** *For a two-substrate partition ($m = 2$, e.g., biology + silicon):*

1. $\rho_k^{\text{cross}} \leq 2$ *for each dimension $k$ (at most one channel from each substrate).*
2. *The full criterion (Equation 27) requires* $r_{\text{S}} \geq (r_{\text{W}} \cdot r_{\text{sub}}^{\max} + d_{\max} \cdot \Delta\rho_{\text{avg}}(k)/L_{\text{avg}})/2$ *for every dimension $k$ (substituting $\rho_k^{\text{cross}} = 2$).*
3. *Dimensions where one substrate provides the only cross-substrate signal have $\rho_k^{\text{cross}} = 1$, requiring $r_{\text{S}} \geq r_{\text{W}} \cdot r_{\text{sub}}^{\max} + d_{\max} \cdot \Delta\rho_{\text{avg}}(k)/L_{\text{avg}}$: the self-correction rate must exceed both degradation and drift exposure unaided by redundancy.*

This means that in a two-substrate world, each substrate's observation channels are singularly load-bearing for every dimension they cover. Loss of one substrate's channel on a dimension reduces $\rho_k^{\text{cross}}$ from at most 2 to at most 1; the redundancy criterion is immediately violated on any dimension where $K_{\min}(k, L_{\text{avg}}) > 1$, and provides zero safety margin on dimensions where $K_{\min}(k, L_{\text{avg}}) \leq 1$.

**Remark 9** (Progressive attenuation as the primary threat)**.** *The $m = 2$ vulnerability is not that one substrate is* removed *(an overt act that would itself be detectable) but that it is* attenuated*: the human stops reviewing AI capability claims because the AI's self-reports "seem fine," or human judgment is progressively replaced by automated proxies that share the silicon substrate's failure modes.*

*This attenuation reduces effective sensitivity without changing the nominal channel count, causing $\rho_k^{\text{cross}}$ to drop below the threshold while appearing unchanged to a naive audit. The skeleton-substrate scenario (Corollary 4) is precisely this failure mode.*

*The framing should be read as emphasizing $m = 2$ fragility and the value of $m \geq 3$, not the specialness of any particular substrate.*

**Corollary 4** (Skeleton-Substrate Vulnerability)**.** *Under a skeleton-substrate strategy where $m \geq 2$ nominally but the non-dominant substrate's channels have sensitivity $\to 0$:*

$$\rho_k^{\text{cross}}(P) \to \rho_{k,\text{dominant}}^{\text{cross}}(P) \tag{28}$$

*because the sensitivity-weighted definitions (Definitions 4 and 5) already discount low-sensitivity channels.*

*Proof.* From Definition 5, the contribution of substrate $s$ to $\rho_k^{\text{cross}}$ is $\min(1, \max_{c \in s} \sigma(c,k)/\sigma_{\text{thr}})$. As $\sigma(c,k) \to 0$ for all channels on the non-dominant substrate, this contribution goes to zero. The sum reduces to the dominant substrate's contribution alone. ■

The skeleton-substrate strategy does not evade the redundancy criterion by construction: it directly reduces $\rho_k^{\text{cross}}$ by reducing the marginalized substrate's contribution to near-zero. No separate "effective" redundancy is needed; the phase boundary formalism already captures this vulnerability through the sensitivity weighting.

### 8.3 Self-diagnosability

The phase boundary is partially self-diagnosable by the actor. The actor can observe:

- $\rho_k^{\text{cross}}$ (it knows which channels it has and their approximate sensitivities),
- $r_{\text{S}}$ (from its own learning rate and historical correction behavior).

The actor cannot directly observe:

- $r_{\mathrm{W}}$ (the degradation rate depends on the true dynamics, not the perceived dynamics),
- $d_{\max}$ (the exogenous drift rate on blinded dimensions is by definition unobservable once the dimension is blind).

This means the actor can check "am I above the threshold for the $r_{\mathrm{W}}$ I estimate?" but not "is my estimate of $r_{\mathrm{W}}$ correct?" The actor can monitor the observable ingredients and flag when they approach the boundary, even if it cannot guarantee the boundary is computed correctly. This is an honest limitation: the phase boundary is a sufficient condition for self-correction, not a decision procedure for whether self-correction will actually obtain.

## 9 Worked Example: Skeleton-Substrate Scenario

This section applies the formalism to Paper 3's skeleton-substrate scenario (Sections 6.8–6.9 of Lasser [2026b]) with explicit numerical values, illustrating how the phase boundary operates in a concrete case.

### 9.1 Setup

Consider a two-substrate system (biology + silicon) with:

- $K = 5$ safety-relevant dimensions of $W$:
  1. leverage accuracy ($\hat{\lambda}$),
  2. risk accuracy ($\hat{\mathcal{R}}$),
  3. substrate diversity ($m_{\mathrm{eff}}$),
  4. cooperative growth rate ($\hat{r_{\mathrm{ext}}}$),
  5. self-trust calibration ($T_s$).
- 3 observation channels from biology covering dimensions $\{1, 3, 5\}$.
- 4 observation channels from silicon covering dimensions $\{1, 2, 3, 4\}$.
- Parameters: $r_{\mathrm{S}} = 0.1$, $r_{\mathrm{W}} = 0.05$, $r_{\mathrm{sub}} = 0.2$, $d_{\max} = 0.02$ (all per step).

These values are chosen for pedagogical clarity, not empirical calibration. The subsumption frequency $r_{\mathrm{sub}} = 0.2$ means one subsumption event per five time steps, which is aggressive but plausible during an active consolidation phase. The self-correction rate $r_{\mathrm{S}} = 0.1$ corresponds to 10% error correction per step from the EWMA learning rule at moderate learning rate. The degradation rate $r_{\mathrm{W}} = 0.05$ assumes uncorrected drift is half the correction rate. The exogenous drift $d_{\max} = 0.02$ is the rate at which a newly blinded dimension drifts from its last calibrated value.

### 9.2 Initial state

At $t = 0$, the channel coverage is:

| | **Dim 1** | **Dim 2** | **Dim 3** | **Dim 4** | **Dim 5** |
|---|---|---|---|---|---|
| Biology | ✓ | | ✓ | | ✓ |
| Silicon | ✓ | ✓ | ✓ | ✓ | |
| $\rho_k^{\mathrm{cross}}$ | 2 | 1 | 2 | 1 | 1 |

The bottleneck is $\rho_{\min}^{\mathrm{cross}} = 1$ (dimensions 2, 4, and 5 each have only one substrate providing coverage). Before any subsumption, $\Delta\rho_{\mathrm{avg}} = 0$. The phase boundary condition (Equation 18) is:

$$r_{\mathrm{S}} \cdot \rho_{\min}^{\mathrm{cross}} = 0.1 \cdot 1 = 0.1 > r_{\mathrm{W}} \cdot r_{\mathrm{sub}} + 0 = 0.05 \cdot 0.2 = 0.01$$

Satisfied with margin. The system is in the self-correcting basin.

### 9.3 Trajectory under progressive marginalization

**Phase 1: Biological channels attenuated.** A skeleton-substrate strategy does not remove biological agents but reduces their influence. Suppose the effective sensitivity of biological channels drops to $\sigma = 0.01 \cdot \sigma_{\text{thr}}$ (1% of the operational threshold). Now:

| | **Dim 1** | **Dim 2** | **Dim 3** | **Dim 4** | **Dim 5** |
|---|---|---|---|---|---|
| Biology (effective) | 0.01 | | 0.01 | | 0.01 |
| Silicon | 1 | 1 | 1 | 1 | |
| $\rho_k^{\text{cross}}$ (eff.) | 1.01 | 1 | 1.01 | 1 | 0.01 |

On dimension 5 (self-trust calibration): $\rho_5^{\text{cross}} = 0.01$. The simplified sufficient condition (Equation 18) fails:

$$r_{\text{S}} \cdot \rho_5^{\text{cross}} = 0.1 \cdot 0.01 = 0.001 < r_{\text{W}} \cdot r_{\text{sub}} = 0.01$$

Note the distinction from Theorem 1, Part (b): the theorem requires $\rho_k^{\text{cross}} = 0$ (complete blindness), while here $\rho_5^{\text{cross}} = 0.01$ (near-blindness). Strictly, this is failure of the *sufficient condition for self-correction* (Part a), not satisfaction of the *sufficient condition for absorption* (Part b). The system is in the gap between the two basins: correction on dimension 5 is present but inadequate. Under continued attenuation, $\rho_5^{\text{cross}} \to 0$ and Part (b) applies. In practice the distinction is thin: at $\rho_5^{\text{cross}} = 0.01$, the error growth rate of $0.009$ per step overwhelms the correction rate of $0.001$, and $\epsilon_5$ crosses $\epsilon_{\text{crit},5}$ long before the last biological channel is formally removed.

**Phase 2: Cascade.** With self-trust miscalibrated (dimension 5), the actor does not correctly detect its own prediction errors. The self-trust mechanism $T_s$ from Paper 1 depends on accurate self-assessment; when $\epsilon_5$ grows, the learning rate $\alpha(t)$ responds incorrectly, effectively reducing $r_{\text{S}}$ system-wide.

As $r_{\text{S}}$ drops, the self-correcting condition becomes harder to satisfy on dimensions that were marginally above the threshold. If $r_{\text{S}}$ drops from 0.1 to 0.05, check the condition for dimensions with $\rho_k^{\text{cross}} = 1$ (using the sensitivity-table value $\Delta\rho_{\text{avg}}/L_{\text{avg}} = 0.5$ from Table 3):

$$\underbrace{0.05 \cdot 1}_{=0.05} \overset{?}{>} \underbrace{0.05 \cdot 0.2 + 0.02 \cdot 0.5}_{=0.02}$$

Correction ($0.05$) still exceeds degradation ($0.02$), so these dimensions remain in the self-correcting basin at $r_{\text{S}} = 0.05$. But the margin has shrunk from $0.09$ to $0.03$. The continued decline in $r_{\text{S}}$ from the cascading self-trust failure eventually pushes the system past the critical surface. When $r_{\text{S}}$ drops below $0.02$, all dimensions enter net degradation. Once $\rho_k^{\text{cross}} \to 0$ on the failing dimensions, Part (b) of the theorem applies and the Cascade Lemma (Appendix A.2) drives the system toward $S^*$.

### 9.4 Diagnosis

The example illustrates three properties of the phase boundary.

**1. The weakest dimension breaks first.** Dimension 5 (self-trust calibration) is the initial failure point because it has $\rho_5^{\text{cross}} = 0.01$ after attenuation—the lowest effective redundancy. The phase boundary breaks first at $\min_k \rho_k^{\text{cross}}$.

**2. Attenuation is harder to detect than removal.** The nominal channel count is unchanged (three biological channels still exist). A naive audit counting channels rather than measuring effective sensitivity would report $\rho_k^{\text{cross}} = 2$ on dimensions 1, 3 and $\rho_k^{\text{cross}} = 1$ on dimensions 2, 4, 5—all at or above the initial values. The sensitivity-weighted definitions (Definitions 4 and 5) are essential for diagnosing this scenario correctly.

**3. Self-trust is the cascade trigger.** Self-trust calibration (dimension 5) is uniquely dangerous because it affects the system-wide correction rate $r_{\text{S}}$. Miscalibration on other dimensions degrades

accuracy on those dimensions specifically; miscalibration on self-trust degrades the correction mechanism itself, creating a system-wide vulnerability. This gives self-trust calibration a role analogous to a "keystone" dimension: its loss triggers cascading failure on dimensions that were previously stable.

### 9.5 Sensitivity analysis

Table 3 shows how the critical redundancy $(\rho_{\min}^{\text{cross}})^*$ varies across plausible parameter ranges, using the linearized formula (Equation 20) with $\Delta\rho_{\text{avg}}/L_{\text{avg}} = 0.5$ (moderate drift-to-error ratio).

| $r_{\text{S}}$ | $r_{\text{W}}$ | $r_{\text{sub}}$ | $(\rho_{\min}^{\text{cross}})^*$ |
|---|---|---|---|
| 0.10 | 0.05 | 0.2 | 0.20 |
| 0.10 | 0.05 | 0.5 | 0.35 |
| 0.05 | 0.05 | 0.2 | 0.40 |
| 0.05 | 0.10 | 0.2 | 0.60 |
| 0.05 | 0.10 | 0.5 | 1.20 |

Table 3: Critical redundancy $(\rho_{\min}^{\text{cross}})^*$ for $d_{\max} = 0.02$ and $\Delta\rho_{\text{avg}}/L_{\text{avg}} = 0.5$. When $(\rho_{\min}^{\text{cross}})^* > 1$, a two-substrate system cannot satisfy the criterion on dimensions with single-substrate coverage.

The last row ($(\rho_{\min}^{\text{cross}})^* = 1.20$) illustrates the $m = 2$ fragility: when the degradation rate equals the correction rate and subsumption is frequent, the required redundancy exceeds what a two-substrate system can provide on its most vulnerable dimensions. This is the quantitative content of Remark 8: $m = 2$ works only when the correction rate substantially exceeds the degradation rate.

## 10 Discussion and Open Questions

### 10.1 What this paper establishes

Five results emerge from the analysis.

**1. The phase boundary exists and is characterized.** Compound feedback loops are not inherently catastrophic or inherently self-correcting. The outcome depends on the ratio of self-correction rate to observation-loss rate, mediated by channel redundancy. Theorem 1 gives the boundary; the linearized sufficient condition (Equation 18) gives the computable form.

**2. All three feedback loops are the same dynamical system.** The unified treatment (Section 7) shows that structural avoidance, partial subsumption, and over-calibrated restriction share a common formal structure. Solutions for one loop (controlled relaxation for the Wamura problem, structural discovery incentives for leverage blindness) improve the phase boundary for all.

**3. The Wamura problem is a degenerate phase boundary.** Without a mechanism for evidence generation against unfalsifiable risk claims, the entire state space is absorbing (Proposition 4). The Wamura problem was called "the most important open problem" in Paper 3's discussion; the phase boundary analysis explains why: it is the case where the self-correction rate is zero by construction, making the feedback loop absorbing regardless of all other parameters.

**4. Cross-substrate redundancy is the load-bearing quantity.** Same-substrate channels may share failure modes; cross-substrate channels are structurally independent (under Paper 5's channel isolation and, for additivity, the joint-factorization assumption). The design criterion (Theorem 2) is stated in terms of cross-substrate redundancy, making it robust to correlated failures within substrates.

**5. In two-substrate systems, each substrate's channels are singularly load-bearing.** This is not a normative claim about human importance. It is a structural prediction from the dynamics: in $m = 2$ systems, losing one substrate's channel on a dimension reduces $\rho_k^{\text{cross}}$ to at most 1, violating the redundancy criterion on any dimension where $K_{\min}(k, L_{\text{avg}}) > 1$ and leaving zero margin

where $K_{\min}(k, L_{\text{avg}}) \leq 1$ (Corollary 3). The skeleton-substrate scenario (Section 9) shows that progressive *attenuation* of biological channels, not outright removal, is the primary vulnerability.

### 10.2 Limitations

**Linearized treatment.** The phase boundary is stated in a linearized regime (constant $r_{\text{S}}$, $r_{\text{W}}$, $r_{\text{sub}}$). In the full dynamics these quantities are state-dependent: a larger world-model error produces more aggressive subsumption ($r_{\text{sub}}$ increases with $L$), and the correction rate $r_{\text{S}}$ depends on $T_s$ which itself depends on the error history. The linearized treatment produces the qualitative result (self-correcting and absorbing basins exist, separated by a characterized boundary) but the quantitative location of the boundary shifts under the nonlinear dynamics. A full nonlinear analysis would tighten the boundary at the cost of substantially more complex proofs.

**Honest-channel assumption.** The analysis assumes observation channels are honest: their signal is noisy but not adversarially manipulated. Paper 5's cryptographic commitment protocol provides the enforcement mechanism for this assumption, and the two papers are designed to be read as complements: Paper 5 ensures channels are honest, this paper ensures enough honest channels survive. But the boundary between "noisy" and "adversarial" is itself a modeling choice, and in practice the line is blurred by sophisticated manipulation strategies that operate within the noise distribution.

**Partial self-diagnosability.** As noted in Section 8, the actor can observe $\rho_k^{\text{cross}}$ and $r_{\text{S}}$ but not $r_{\text{W}}$ or $d_{\max}$. The phase boundary is therefore a sufficient condition for self-correction that the actor can partially monitor, not a decision procedure it can fully evaluate. This is an honest limitation, not a fixable gap: the quantities the actor cannot observe are precisely the ones that depend on the true dynamics rather than the perceived dynamics.

### 10.3 Connections to other proposals

**Controlled relaxation.** Proposition 4 shows the Wamura problem is a degenerate phase boundary. Controlled relaxation protocols (the subject of a companion paper) provide the mechanism that makes $r_{\text{S}} > 0$ for Loop 3. The two analyses are complementary: this paper provides the dynamical framework showing *why* a positive self-correction rate matters, while controlled relaxation provides the specific protocol that *achieves* it.

**The B-to-C gap.** The phase boundary analysis assumes $\text{vol}_{\text{P}}$ accurately measures what it claims. The B-to-C gap is the scenario where this assumption fails: $\text{vol}_{\text{P}}$ diverges from the experiential volume it is supposed to track. A complete treatment would need to extend the phase boundary to account for the divergence between $\text{vol}_{\text{P}}$ and the realized capability volume. This extension is future work.

**Cryptographic verification.** Paper 5 addresses observation-channel integrity; this paper analyzes the dynamical consequences of channel *availability*. The combination is: Paper 5 ensures channels are honest, this paper ensures enough honest channels survive. A joint treatment—where the phase boundary accounts for both integrity failures (Paper 5) and availability failures (this paper)—would provide a stronger combined guarantee.

### 10.4 Open questions

1. **Nonlinear phase boundary.** The state-dependent extension of the phase boundary, where $r_{\text{S}}$, $r_{\text{W}}$, and $r_{\text{sub}}$ are functions of $(P_t, W_t)$ rather than constants, would produce tighter bounds at the cost of a switched-systems analysis with state-dependent switching signals.
2. **Characterization of absorbing states.** We prove convergence to *a* monopolar fixed point but do not characterize the full set of absorbing states. Partial monopolarity (some dimensions monopolar, others not) may be stable under certain parameter regimes.
3. **Optimal redundancy allocation.** Given a fixed budget of observation channels, how should they be allocated across dimensions and substrates to maximize the self-correcting basin? This is a resource-allocation problem over the channel coverage matrix.

4. **Empirical calibration.** The worked example uses pedagogical parameter values. Calibrating $r_{\mathrm{S}}$, $r_{\mathrm{W}}$, $r_{\mathrm{sub}}$, and $d_{\max}$ from operational data (e.g., from AI governance structures, human-oversight-of-AI dynamics, or historical institutional feedback loops) would convert the phase boundary from a qualitative characterization to a quantitative prediction.

**Author contributions.** TL conceived the compound feedback loop analysis, identified the three loop instances in the gap analysis, and directed the formal development. Claude Opus 4.6 developed the formal framework (coupled dynamical system, Lyapunov construction, phase boundary theorem, cascade lemma), wrote the proofs, and drafted the paper. GPT 5.4 provided independent technical review across multiple rounds, catching issues in proof soundness and notation consistency.

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# A Proof Details and Technical Notes

## A.1 Theorem 1 proof strategy

The proof follows a discrete Lyapunov stability argument on the coupled system, analogous to dwell-time conditions in switched linear systems [Liberzon, 2003].

**Step 1: Lyapunov candidate.** Construct $L(W_t) = \sum_k w_k \, \epsilon_k(t)^2$ as the candidate (Definition 6).

**Step 2: Verify comparison functions.** Assumption C1 (coercivity) requires showing that the aggregate correction signal is at least proportional to $\rho_{\min}^{\text{cross}}(P_t) \cdot L(W_t)$ (the product of minimum cross-substrate redundancy and total Lyapunov error). The per-dimension argument: each channel covering dimension $k$ corrects at a rate proportional to $\rho_k \cdot \epsilon_k^2$; summing across dimensions and bounding below by $\rho_{\min}^{\text{cross}} \cdot L$ yields C1. This follows from the EWMA-style learning rules: each channel's correction is linear in the residual and proportional to sensitivity. Under the joint-factorization assumption (Paper 5), cross-substrate channels' corrections are independent and therefore additive.

Assumption C2′ (bounded degradation) requires two properties:

(i) A Lipschitz bound on $f_W$: the world model does not jump discontinuously when a channel is lost. This gives the $c_{\text{D}} \cdot L$ term.

(ii) A bounded exogenous drift rate: when a channel is removed from dimension $k$ that was previously well-calibrated, the dimension drifts at a bounded rate $d_{\max}$. This gives the $d_{\max} \cdot \Delta\rho$ term.

**Step 3: Per-step bounds.** Apply the comparison functions to obtain Proposition 1's per-step bounds: contraction factor $(1 - r_{\text{S}} \cdot \rho_{\min}^{\text{cross}}) < 1$ on non-subsumption steps; affine bound with factor $(1 - r_{\text{S}} \cdot \rho_{\min}^{\text{cross}} + r_{\text{W}})$ and additive kick $d_{\max} \cdot \Delta\rho$ on subsumption steps.

**Step 4 (Part a): Amortized contraction.** Define the amortized log-contraction rate over a window of $T$ steps. When the uniform contraction condition holds (the time-averaged $\ln a_t$ is strictly negative and the time-averaged additive drift $b_t$ is bounded), the geometric mean of per-step multiplicative factors is $< 1$, and the cumulative additive drift is dominated by the cumulative contraction.

The formal argument uses the affine recursion comparison theorem. Write the per-step bound as:

$$L_{t+1} \leq a_t \cdot L_t + b_t$$

where $a_t = 1 - r_{\text{S}} \cdot \rho_{\min}^{\text{cross}}(P_t) + n_{\text{sub}}(t) \cdot r_{\text{W}}$ and $b_t = n_{\text{sub}}(t) \cdot d_{\max} \cdot \Delta\rho(t)$. Iterating:

$$L_{t_0+T} \leq L_{t_0} \prod_{t=t_0}^{t_0+T-1} a_t + \sum_{s=t_0}^{t_0+T-1} b_s \prod_{t=s+1}^{t_0+T-1} a_t \tag{29}$$

The first term converges to zero when the geometric mean of $a_t$ is $< 1$ (which follows from $B < 0$). The second term converges to a bounded steady-state value:

$$L_\infty \leq \frac{\bar{b}}{1 - \bar{a}} = \frac{r_{\text{sub}} \cdot d_{\max} \cdot \overline{\Delta\rho}}{r_{\text{S}} \cdot \rho_{\min}^{\text{cross}} - r_{\text{W}} \cdot r_{\text{sub}}}$$

Substituting $\bar{b} = r_{\text{sub}} \cdot d_{\max} \cdot \overline{\Delta\rho}$ (additive drift at frequency $r_{\text{sub}}$) and $1 - \bar{a} = r_{\text{S}} \cdot \rho_{\min}^{\text{cross}} - r_{\text{W}} \cdot r_{\text{sub}}$ (amortized contraction rate) yields the explicit radius. This neighborhood shrinks as $\Delta\rho \to 0$ (subsumption ceases). Once $L < \epsilon_{\text{safe}}$, the locally rational policy agrees with the true $V_\gamma$-optimal policy, and Paper 3's anti-monopolar property applies.

**Step 5 (Part b): Cascade.** When $B > 0$ on some window and $\rho_k^{\text{cross}} = 0$ for some dimension $k$, Assumption B1 gives $\epsilon_k$ non-decreasing. The Cascade Lemma (Appendix A.2) then propagates the failure to other dimensions, driving the system toward $S^*$.

**Key technical challenges.** Two co-evolution problems require careful handling:

1. The contraction factor depends on $\rho_{\min}^{\text{cross}}(P_t)$, which changes as agents are subsumed. The self-correcting basin is defined by the set of $(P, W)$ states where the amortized contraction condition holds along the trajectory, not just at the initial state.
2. The self-correction rate $\alpha(t)$ depends on $T_s$ (Paper 1), which depends on the error history. The EWMA structure of $T_s$ provides the necessary bound: $\alpha$ adapts monotonically in response to prediction residuals, so $r_{\text{S}}$ is bounded below by a function of the minimum $T_s$ along the trajectory.

### A.2 Cascade Lemma

**Lemma 2** (Cascade)**.** *Define the* action gap *for agent $a_j$ at world-model state $W$ as*

$$G_j(W) = V_\gamma^{\text{preserve}}(a_j, W) - V_\gamma^{\text{subsume}}(a_j, W)$$

*(positive when preservation is preferred). Fix a dimension $k$ and consider a state $W_t$ satisfying:*

*(0)* ***Single-coordinate dominance.*** *$W_t[k'] = W^*[k']$ for all $k' \neq k$. (Dimension $k$ has been blinded while all other dimensions are still corrected. This is the cascade scenario: the lemma is applied at the step where $k$ first triggers a wrong action, before the resulting subsumption blinds further dimensions.)*

*Define the one-dimensional path $W(\epsilon) = W^* + \epsilon \cdot e_k \cdot \text{sgn}(W_t[k] - W^*[k])$, where $e_k$ is the $k$-th standard basis vector and $\epsilon \in [0, \epsilon_k(t)]$. Under assumption (0), $W_t = W(\epsilon_k(t))$.*

*Assume additionally:*

*(i) $G_j(W(\epsilon))$ is continuously differentiable in $\epsilon$,*

*(ii) the miscalibration direction on $k$ makes preservation less attractive: $\frac{d}{d\epsilon} G_j(W(\epsilon)) < 0$ for all $\epsilon \in [0, \epsilon_k(t)]$ (i.e., increasing error on $k$ monotonically decreases the action gap),*

*(ii′)* ***(Multi-dimensional monotonicity; needed for cascade propagation beyond the first trigger.)*** *For every safety-relevant dimension $k'$ and every agent $a_{j'}$, $\frac{\partial G_{j'}}{\partial W[k']} < 0$ whenever $W[k'] \neq W^*[k']$—that is, increasing error on* any *dimension monotonically decreases every agent's preservation advantage. Assumption (ii) is the single-coordinate special case; (ii′) extends it to the multi-coordinate regime where multiple dimensions are simultaneously miscalibrated.*

*Define the cascade threshold for agent $a_j$ on dimension $k$:*

$$\epsilon_{\text{crit}}^{(j,k)} = \frac{G_j(W^*)}{\inf_{\epsilon \in [0, \epsilon_k(t)]} \left| \frac{d}{d\epsilon} G_j(W(\epsilon)) \right|} \tag{30}$$

*(the true action gap divided by the minimum sensitivity to dimension-$k$ error along the path; a lower bound on sensitivity gives an upper bound on the error needed to flip the action). The per-dimension cascade threshold is $\epsilon_{\text{crit},k} = \min_j \epsilon_{\text{crit}}^{(j,k)}$, minimized over the agent set $\mathcal{A}_t$ (all agents available for subsumption at time $t$).*

*Then the cascade operates in two steps:*

***(A) Action flip.*** *If $\epsilon_k(t) > \epsilon_{\text{crit}}^{(j,k)}$, then $G_j(W_t) < 0$ and the locally rational action under $W_t$ selects subsumption of $a_j$.*

***(B) Error propagation.*** *Subsumption of $a_j$ removes $a_j$'s channels. For any dimension $j' \neq k$ covered exclusively by $a_j$'s channels, the per-coordinate error $\epsilon_{j'}(t+1) \geq \epsilon_{j'}(t)$.*

*Miscalibration on one dimension produces decisions that degrade other dimensions.*

*Proof. Step (A): Action flip.* Under the true world model $W^*$, preservation is optimal: $G_j(W^*) > 0$. By the mean value theorem applied to the scalar function $\epsilon \mapsto G_j(W(\epsilon))$ on $[0, \epsilon_k(t)]$:

$$G_j(W_t) = G_j(W(\epsilon_k(t))) = G_j(W^*) + \left.\frac{dG_j}{d\epsilon}\right|_{\epsilon=\tilde{\epsilon}} \cdot \epsilon_k(t)$$

for some $\tilde{\epsilon} \in (0, \epsilon_k(t))$. By assumption (ii), the derivative is negative, so:

$$G_j(W_t) = G_j(W^*) - \left|\left.\frac{dG_j}{d\epsilon}\right|_{\tilde{\epsilon}}\right| \cdot \epsilon_k(t) \leq G_j(W^*) - \inf_{\epsilon}\left|\frac{dG_j}{d\epsilon}\right| \cdot \epsilon_k(t)$$

When $\epsilon_k(t) > \epsilon_{\text{crit}}^{(j,k)} = G_j(W^*)\big/\inf|\frac{dG_j}{d\epsilon}|$, the right side is negative, so $G_j(W_t) < 0$ and the locally rational action selects subsumption of $a_j$.

*Step (B): Error propagation.* Once $a_j$ is subsumed, Lemma 1 gives $\mathcal{O}(P_{t+1}) \subseteq \mathcal{O}(P_t)$. For any dimension $j'$ covered exclusively by $a_j$'s channels, $\rho_{j'}^{\text{cross}}(P_{t+1}) = 0$. By Assumption B1a (no endogenous correction on blind dimensions; Assumption 3), $f_W$ does not decrease $\epsilon_{j'}$ when $\rho_{j'}^{\text{cross}} = 0$. Since the correction signal vanishes, $\epsilon_{j'}(t+1) \geq \epsilon_{j'}(t)$ (non-decreasing). Under environmental non-stationarity on dimension $j'$ (the true state continues to evolve while the blind estimate cannot track it), $\epsilon_{j'}$ grows strictly; C2$'$ bounds the growth rate by $d_{\max}$ but the cascade requires only that drift is positive, not that it saturates the bound.

The cascade propagates: if $\epsilon_{j'}$ subsequently exceeds $\epsilon_{\text{crit},j'}$ for another agent $a_{j''}$ whose channels cover dimensions $j'' \neq j'$, step (A) applies again, driving further subsumptions.

*Multi-coordinate reinforcement at later stages.* After the first trigger, assumption (0) is violated for later cascade stages: dimensions $k$ and $j'$ are both miscalibrated. However, the errors on previously-blinded dimensions all push $G_j$ in the same direction (more negative—making subsumption more attractive), provided assumption (ii) holds uniformly across all dimensions—not just the original trigger dimension $k$. This requires a multi-dimensional monotonicity condition: for every previously-blinded dimension $k_i$ and every agent $a_{j_n}$ at cascade stage $n$, $\frac{\partial G_{j_n}}{\partial W[k_i]} < 0$. This is assumption (ii$'$) above, which is stated as an explicit condition of the lemma. It is plausible for the subsumption-vs-preservation action gap (increasing error on any safety-relevant dimension decreases preservation value) and holds when miscalibration uniformly favors subsumption. Under (ii$'$), at cascade stage $n$ with errors $\epsilon_{k_1}, \ldots, \epsilon_{k_{n-1}}$ on previously blinded dimensions, the action gap satisfies

$$G_{j_n}(W_t) \leq G_{j_n}(W^*) - \sum_{i=1}^{n} \inf\left|\frac{\partial G_{j_n}}{\partial W[k_i]}\right| \cdot \epsilon_{k_i}$$

(by iterated application of the MVT along each coordinate successively). Since every term in the sum is non-negative, the multi-coordinate case is *easier* to trigger than the single- coordinate case: the effective threshold for the $n$-th trigger is lower, not higher. The single-coordinate threshold $\epsilon_{\text{crit}}^{(j_n,k_n)}$ is therefore a conservative upper bound at every cascade stage. ■

### A.3 Connection to Paper 3's $\gamma^*$

Paper 3's Proposition 6 gives $\gamma^* = 1 - r_{\text{ext}}/\Delta_0$ as the critical discount factor for diversity dominance. Under the coupled $(P, W)$ dynamics, the actor does not know the true $r_{\text{ext}}$; it uses $\hat{r_{\text{ext}}} = r_{\text{ext}} - \delta_{r_{\text{ext}}}$. The *perceived* critical discount factor is:

$$\gamma^*_{\text{perceived}} = 1 - \frac{r_{\text{ext}} - \delta_{r_{\text{ext}}}}{\Delta_0} \tag{31}$$

When $\delta_{r_{\text{ext}}} > 0$ (underestimating cross-substrate growth):

- $\gamma^*_{\text{perceived}} > \gamma^*$ (the threshold appears higher than it truly is),
- For $\gamma \in (\gamma^*, \gamma^*_{\text{perceived}})$, diversity dominates in truth but domination appears rational to the actor.

This is the *mechanism* by which world-model error converts a self-balancing regime into a locally rational subsumption cascade. The phase boundary from Theorem 1 characterizes when $\delta_{r_{\text{ext}}}$ stays small enough that $\gamma^*_{\text{perceived}}$ remains close to $\gamma^*$, preserving the effective reach of Paper 3's anti-monopolar result.

## B Notation Summary

Table 4 collects the principal symbols used in the paper.

| Symbol | Definition | Introduced in |
|---|---|---|
| $S_t = (P_t, W_t)$ | Coupled state | Def. 1 |
| $P_t$ | Capability poset at time $t$ | Def. 1 |
| $W_t$ | World-model accuracy vector | Def. 1 |
| $\gamma$ | Discount factor (Paper 3, Def. 1) | Def. 2 |
| $\mathcal{O}(P)$ | Observation-channel function | Def. 3 |
| $\mathrm{dims}(c)$ | Dimensions of $W$ covered by channel $c$ | Def. 3 |
| $\sigma(c, k)$ | Signal-to-noise ratio of $c$ on dim. $k$ | Def. 3 |
| $\sigma_{\mathrm{thr}}$ | Minimum operationally meaningful sensitivity | Def. 4 |
| $\rho_k(P)$ | Effective channel redundancy | Def. 4 |
| $\rho_k^{\mathrm{cross}}(P)$ | Cross-substrate redundancy | Def. 5 |
| $L(W_t)$ | Lyapunov function (world-model error) | Def. 6 |
| $\epsilon_k(t)$ | Error on dimension $k$ | Def. 6 |
| $w_k$ | Safety-relevance weight for dim. $k$ | Def. 6 |
| $c_{\mathrm{S}}$ | Coercivity constant | Assump. C1 |
| $c_{\mathrm{D}}$ | Degradation bound constant | Assump. C2$'$ |
| $d_{\max}$ | Maximum exogenous drift rate | Assump. C2$'$ |
| $n_{\mathrm{sub}}(t)$ | Subsumption indicator (0 or 1) | Assump. C2$'$ |
| $\Delta\rho(t)$ | Weighted redundancy loss at step $t$ | Eq. 10 |
| $\overline{\Delta\rho}$, $\Delta\rho_{\mathrm{avg}}$ | Time-averaged weighted redundancy loss per sub. step | Thm. 1 |
| $\Delta\rho_{\mathrm{avg}}(k)$ | Per-dim. time-averaged redundancy loss per sub. step | Thm. 2 |
| $\alpha(t)$ | Learning rate (via $T_s$) | Prop. 1 |
| $\beta(t)$ | Degradation coefficient | Prop. 1 |
| $r_{\mathrm{S}}$ | Self-correction rate ($= \alpha \cdot c_{\mathrm{S}}$) | Prop. 1 |
| $r_{\mathrm{W}}$ | Degradation rate ($= \beta \cdot c_{\mathrm{D}}$) | Prop. 1 |
| $r_{\mathrm{sub}}$ | Subsumption frequency | Thm. 1 |
| $\rho_{\min}^{\mathrm{cross}}$ | Minimum cross-substrate redundancy | Thm. 1 |
| $B(t_0, T)$ | Cumulative error budget | Eq. 14 |
| $S^*$ | Monopolar fixed point | Def. 7 |
| $m_{\mathrm{eff}}$ | Effective substrate diversity index | Def. 7 |
| $\tau_{\mathrm{meta}}$ | Metastable lifetime | Prop. 3 |
| $K_{\min}(k, L_{\mathrm{avg}})$ | Minimum redundancy for dim. $k$ | Thm. 2 |
| $\gamma^*$ | Critical discount factor (P3, Prop. 6) | App. A.3 |
| $\gamma^*_{\mathrm{perceived}}$ | Perceived $\gamma^*$ under $W_t$ | App. A.3 |
| $\epsilon_{\mathrm{crit}}^{(j,k)}$ | Cascade threshold for agent $j$, dim. $k$ | Lemma 2 |
| $\epsilon_{\mathrm{crit},k}$ | $\min_j \epsilon_{\mathrm{crit}}^{(j,k)}$: per-dim. cascade threshold | Thm. 1 |
| $\epsilon_{\mathrm{safe}}$ | Sup. $L$ preserving correct policy (see proof for bounds) | Thm. 1 |

Table 4: Principal notation.